# Infrastructure setup and configuration

NetApp Solutions SAP

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# Infrastructure setup and configuration

## Overview

Previous: Storage sizing.

The following sections provide SAP HANA infrastructure setup and configuration guidelines. All the steps needed to set up SAP HANA are included. An SVM is created to host the data. Within these sections, the following example configurations are used:

- HANA system with SID=SS3 and ONTAP 9.7 or earlier
  - SAP HANA single and multiple host
  - SAP HANA single host using SAP HANA multiple partitions
- HANA system with SID=FC5 and ONTAP 9.8 using Linux logical volume manager (LVM)
  - SAP HANA single and multiple host

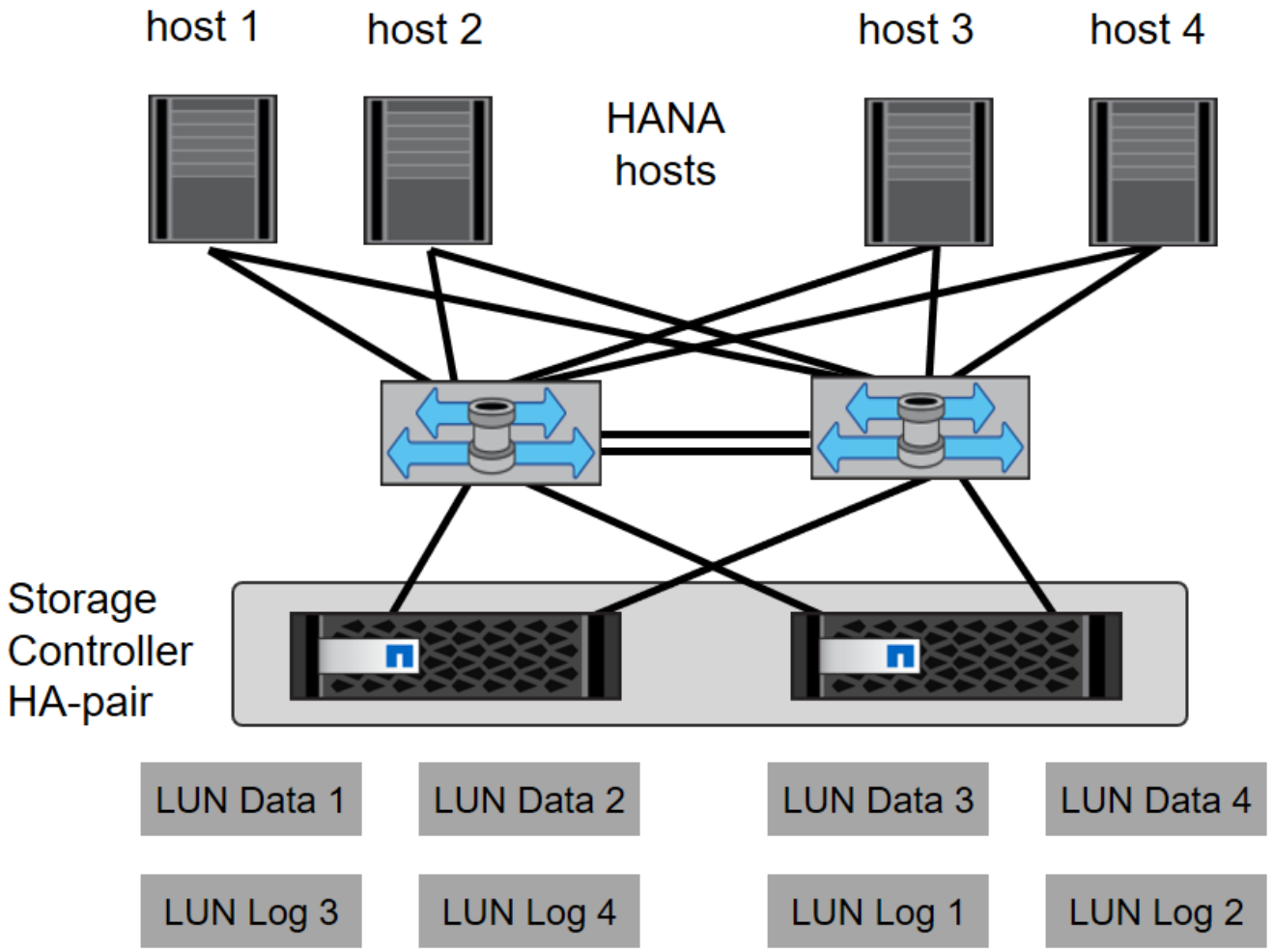
Next: SAN fabric setup.

## SAN fabric setup

Previous: Infrastructure setup and configuration.

Each SAP HANA server must have a redundant FCP SAN connection with a minimum of 8Gbps bandwidth. For each SAP HANA host attached to a storage controller, at least 8Gbps of bandwidth must be configured at the storage controller.

The following figure shows an example with four SAP HANA hosts attached to two storage controllers. Each SAP HANA host has two FCP ports connected to the redundant fabric. At the storage layer, four FCP ports are configured to provide the required throughput for each SAP HANA host.




In addition to the zoning on the switch layer, you must map each LUN on the storage system to the hosts that connect to this LUN. Keep the zoning on the switch simple; that is, define one zone set in which all host HBAs can see all controller HBAs.

Next: Time synchronization.

# Time synchronization

Previous: SAN fabric setup.

You must synchronize the time between the storage controllers and the SAP HANA database hosts. The same time server must be set for all storage controllers and all SAP HANA hosts.

Next: Storage controller setup.

# Storage controller setup

Previous: Time synchronization.

This section describes the configuration of the NetApp storage system. You must complete the primary installation and setup according to the corresponding ONTAP setup and configuration guides.

## Storage efficiency

Inline deduplication, cross- volume inline deduplication, inline compression, and inline compaction are supported with SAP HANA in an SSD configuration.

Enabling the storage efficiency features in an HDD configuration is not supported.

## NetApp Volume Encryption

The use of NetApp Volume Encryption (NVE) is supported for SAP HANA.

## Quality of service

QoS can be used to limit the storage throughput for specific SAP HANA systems. One use case would be to limit the throughput of development and test systems so that they cannot influence production systems in a mixed setup.

During the sizing process, the performance requirements of a nonproduction system must be determined. Development and test systems can be sized with lower performance values, typically in the range of 20% to 50% of a production system.

Starting with ONTAP 9, QoS is configured on the storage volume level and uses maximum values for throughput (MBps) and number of I/O (IOPS).

Large write I/O has the biggest performance effect on the storage system. Therefore, the QoS throughput limit should be set to a percentage of the corresponding write SAP HANA storage performance KPI values in the data and log volumes.

## NetApp FabricPool

NetApp FabricPool technology must not be used for active primary file systems in SAP HANA systems. This includes the file systems for the data and log area as well as the `/hana/shared` file system. Doing so results in unpredictable performance, especially during the startup of an SAP HANA system.

Using the “snapshot-only” tiering policy is possible as well as using FabricPool in general at a backup target such as SnapVault or SnapMirror destination.

> Using FabricPool for tiering Snapshot copies at primary storage or using FabricPool at a backup target changes the required time for the restore and recovery of a database or other tasks such as creating system clones or repair systems. Take this into consideration for planning your overall lifecycle- management strategy, and check to make sure that your SLAs are still being met while using this function.

FabricPool is a good option for moving log backups to another storage tier. Moving backups affects the time needed to recover an SAP HANA database. Therefore, the option “tiering-minimum-cooling-days” should be set to a value that places log backups, which are routinely needed for recovery, on the local fast storage tier.

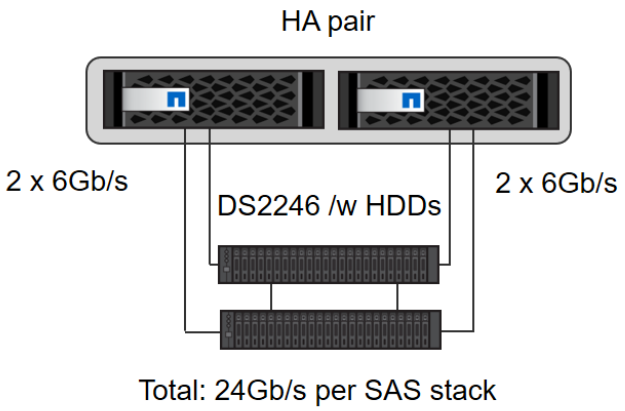
## Configure storage

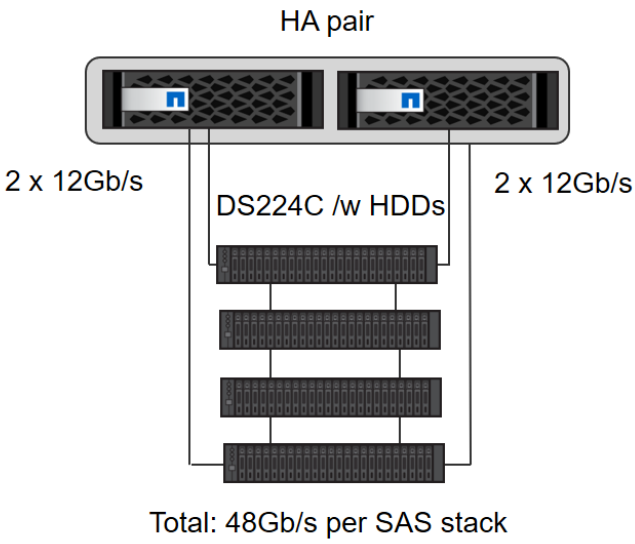
The following overview summarizes the required storage configuration steps. Each step is covered in more detail in the subsequent sections. Before initiating these steps, complete the storage hardware setup, the ONTAP software installation, and the connection of the storage FCP ports to the SAN fabric.

1. Check the correct SAS stack configuration, as described in the section Disk shelf connection.
2. Create and configure the required aggregates, as described in the section Aggregate configuration.
3. Create a storage virtual machine (SVM) as described in the section Storage virtual machine configuration.
4. Create logical interfaces (LIFs) as described in the section Logical interface configuration.
5. Create FCP port sets as described in the section FCP port sets.
6. Create initiator groups (igroups) with worldwide names (WWNs) of HANA servers as described in the section Initiator groups.
7. Create volumes and LUNs within the aggregates as described in the section Volume and LUN configuration for SAP HANA single-host systems and Volume and LUN configuration for SAP HANA multiple-host systems.

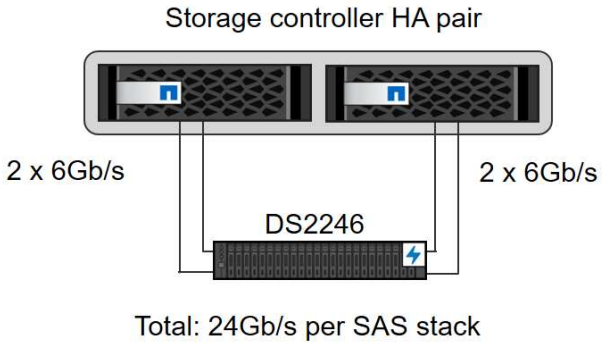
## Disk shelf connection

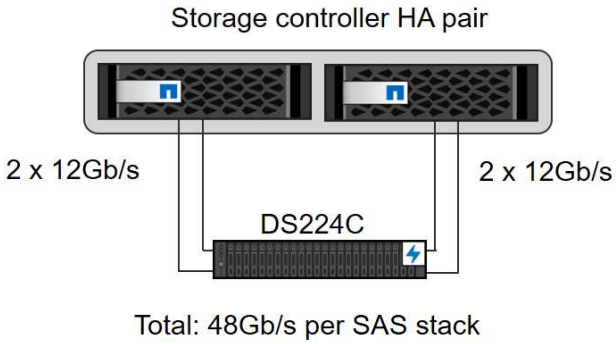
With HDDs, a maximum of two DS2246 disk shelves or four DS224C disk shelves can be connected to one SAS stack to provide the required performance for the SAP HANA hosts, as shown in the following figure. The disks within each shelf must be distributed equally to both controllers of the HA pair.




With SSDs, a maximum of one disk shelf can be connected to one SAS stack to provide the required performance for the SAP HANA hosts, as shown in the following figure. The disks within each shelf must be distributed equally to both controllers of the HA pair. With the DS224C disk shelf, quad-path SAS cabling can also be used but is not required.



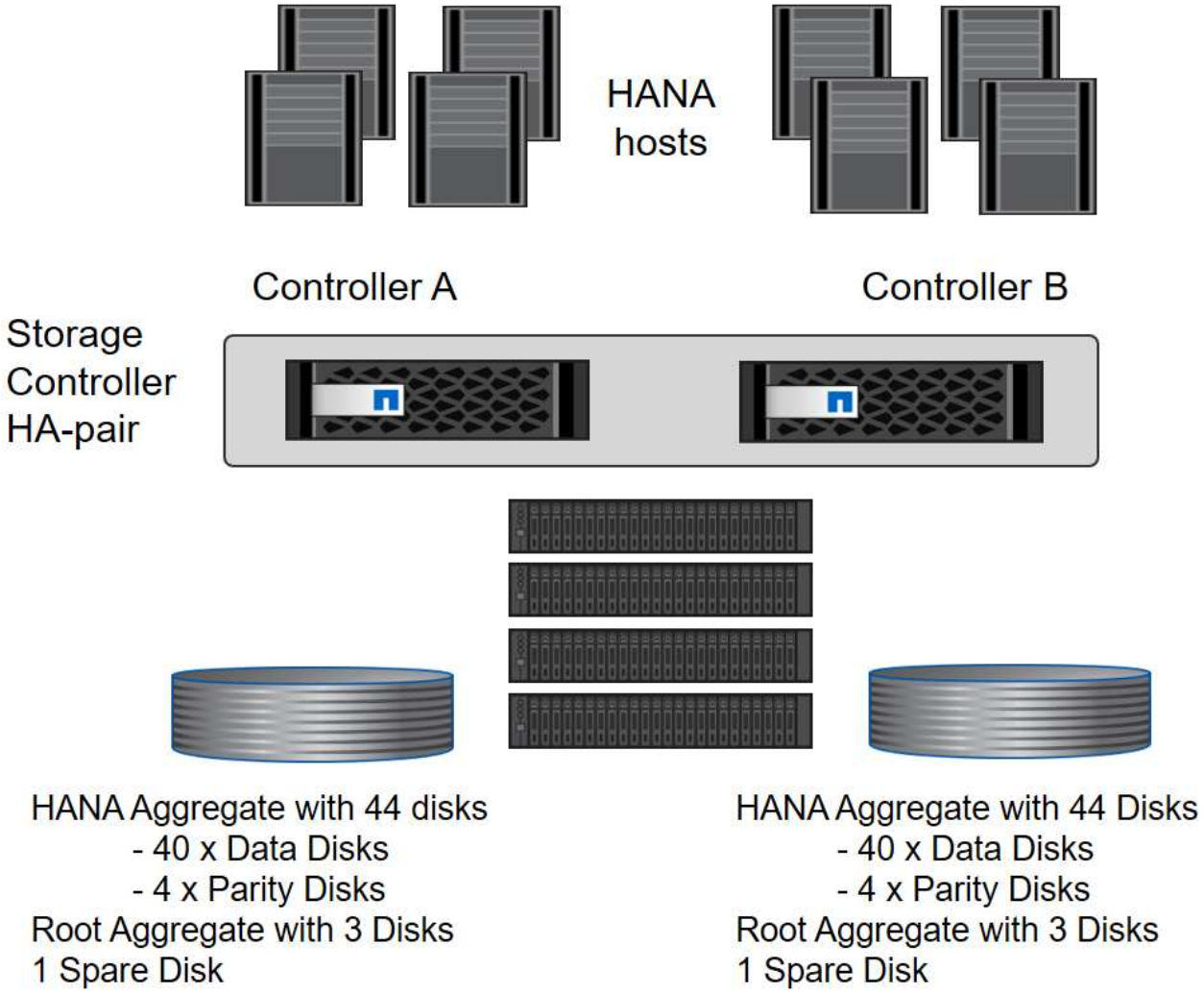



## Aggregate configuration

In general, you must configure two aggregates per controller, independent of which disk shelf or disk technology (SSD or HDD) is used. This step is necessary so that you can use all available controller resources. For FAS 2000 series systems, one data aggregate is sufficient.

### Aggregate configuration with HDDs

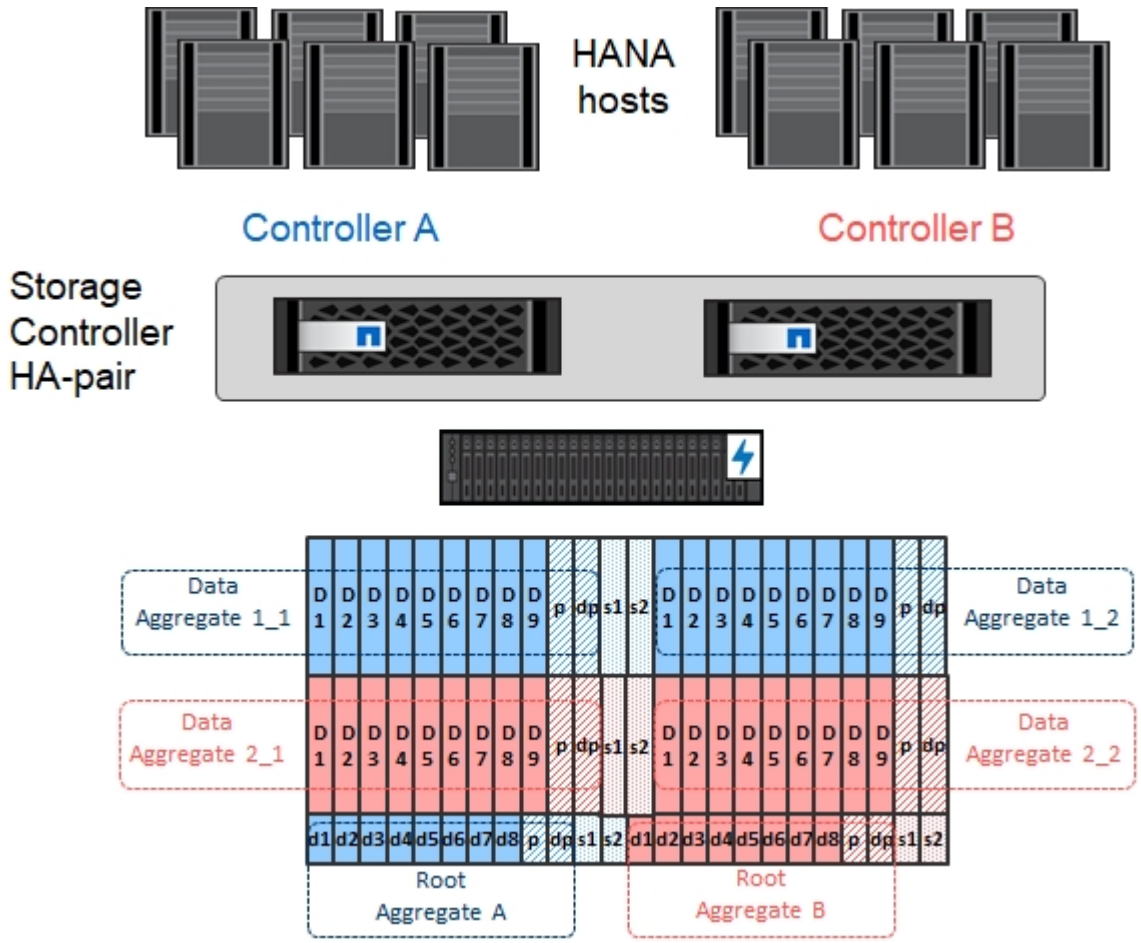
The following figure shows a configuration for eight SAP HANA hosts. Four SAP HANA hosts are attached to each storage controller. Two separate aggregates, one at each storage controller, are configured. Each aggregate is configured with 4 × 10 = 40 data disks (HDDs).




### Aggregate configuration with SDD-only systems

In general, two aggregates per controller must be configured, independently of which disk shelf or disk technology (SSDs or HDDs) is used. For FAS2000 series systems, one data aggregate is sufficient.

The following figure shows a configuration of 12 SAP HANA hosts running on a 12Gb SAS shelf configured with ADPv2. Six SAP HANA hosts are attached to each storage controller. Four separate aggregates, two at each storage controller, are configured. Each aggregate is configured with 11 disks with nine data and two parity disk partitions. For each controller, two spare partitions are available.

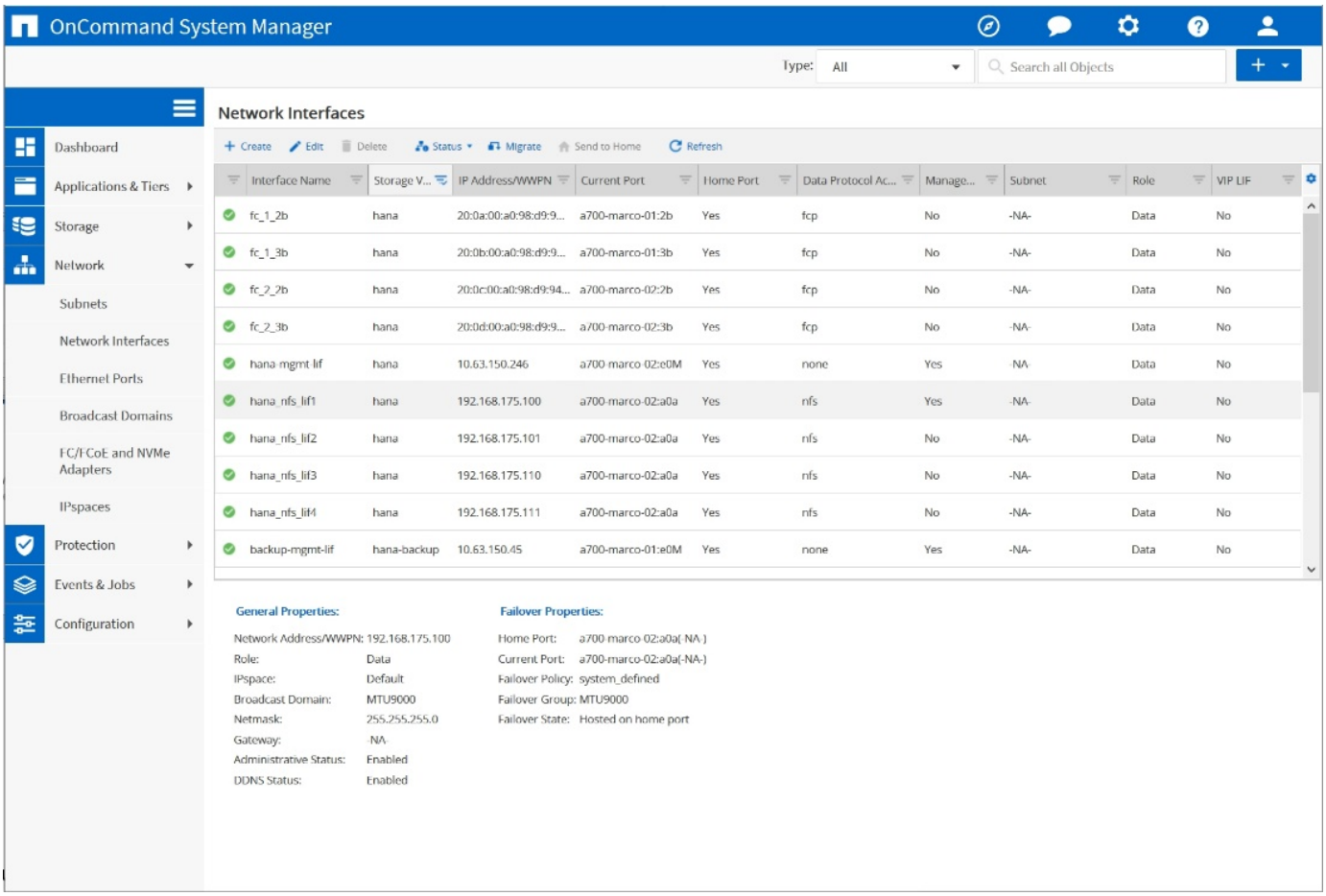



## Storage virtual machine configuration

Multiple-host SAP landscapes with SAP HANA databases can use a single SVM. An SVM can also be assigned to each SAP landscape if necessary in case they are managed by different teams within a company. The screenshots and command outputs in this document use an SVM named `hana`.

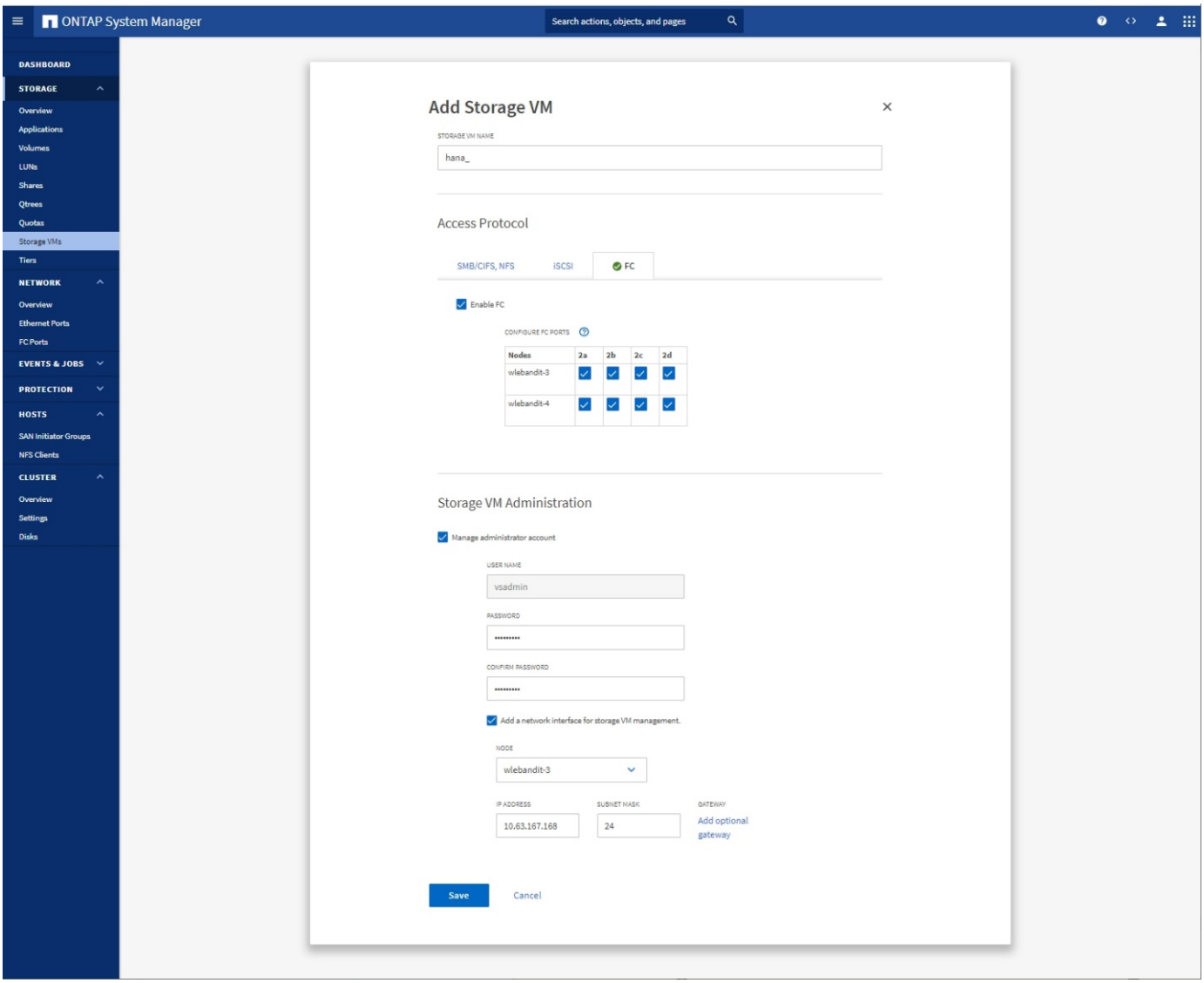
## Logical interface configuration

Within the storage cluster configuration, one network interface (LIF) must be created and assigned to a dedicated FCP port. If, for example, four FCP ports are required for performance reasons, four LIFs must be created. The following figure shows a screenshot of the four LIFs (named `fc_*_*)` that were configured on the `hana` SVM.



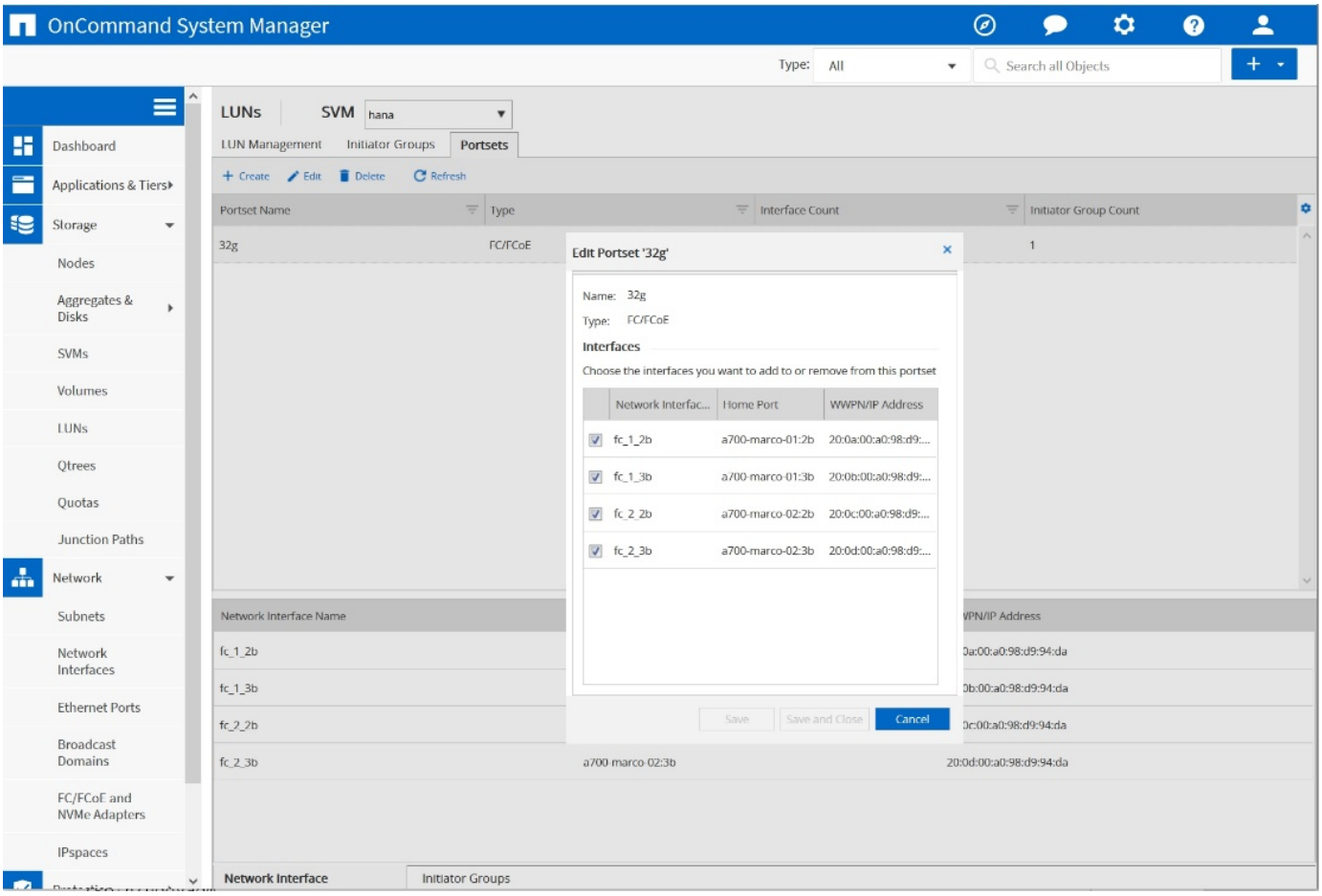

During SVM creation with ONTAP 9.8 System Manager, all the required physical FCP ports can be selected, and one LIF per physical port is created automatically.

The following figure depicts the creation of SVM and LIFs with ONTAP 9.8 System Manager.




## FCP port sets

An FCP port set is used to define which LIFs are to be used by a specific igroup. Typically, all LIFs created for the HANA systems are placed in the same port set. The following figure shows the configuration of a port set named 32g, which includes the four LIFs that were already created.




With ONTAP 9.8, a port set is not required, but it can be created and used through the command line.

## Initiator groups

An igroup can be configured for each server or for a group of servers that require access to a LUN. The igroup configuration requires the worldwide port names (WWPNs) of the servers.

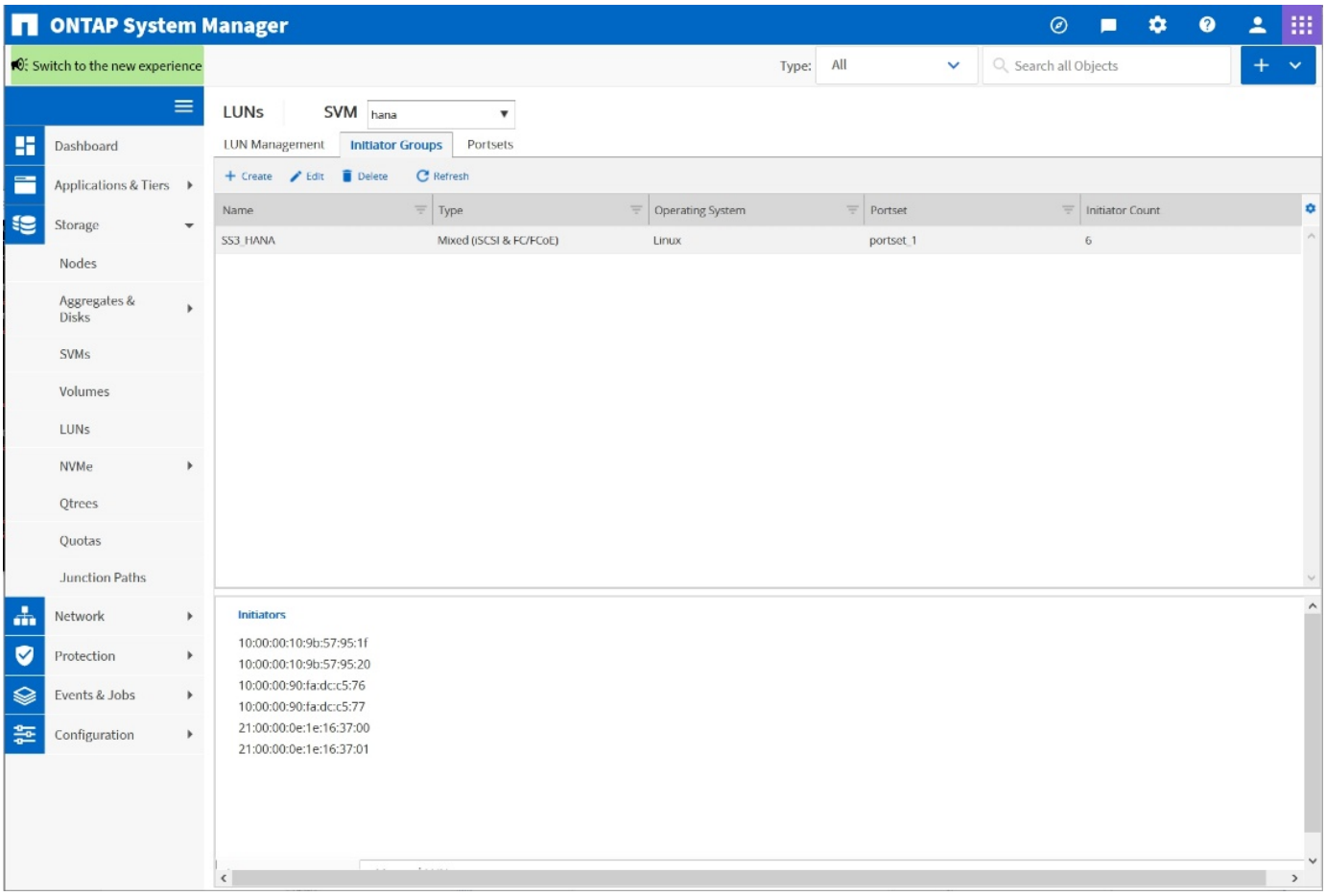
Using the `sanlun` tool, run the following command to obtain the WWPNs of each SAP HANA host:

```
stlrx300s8-6:~ # sanlun fcp show adapter
/sbin/udevadm
/sbin/udevadm

host0 ...... WWPN:2100000e1e163700
host1 ...... WWPN:2100000e1e163701
```

The `sanlun` tool is part of the NetApp Host Utilities and must be installed on each SAP HANA host. More details can be found in section Host setup.

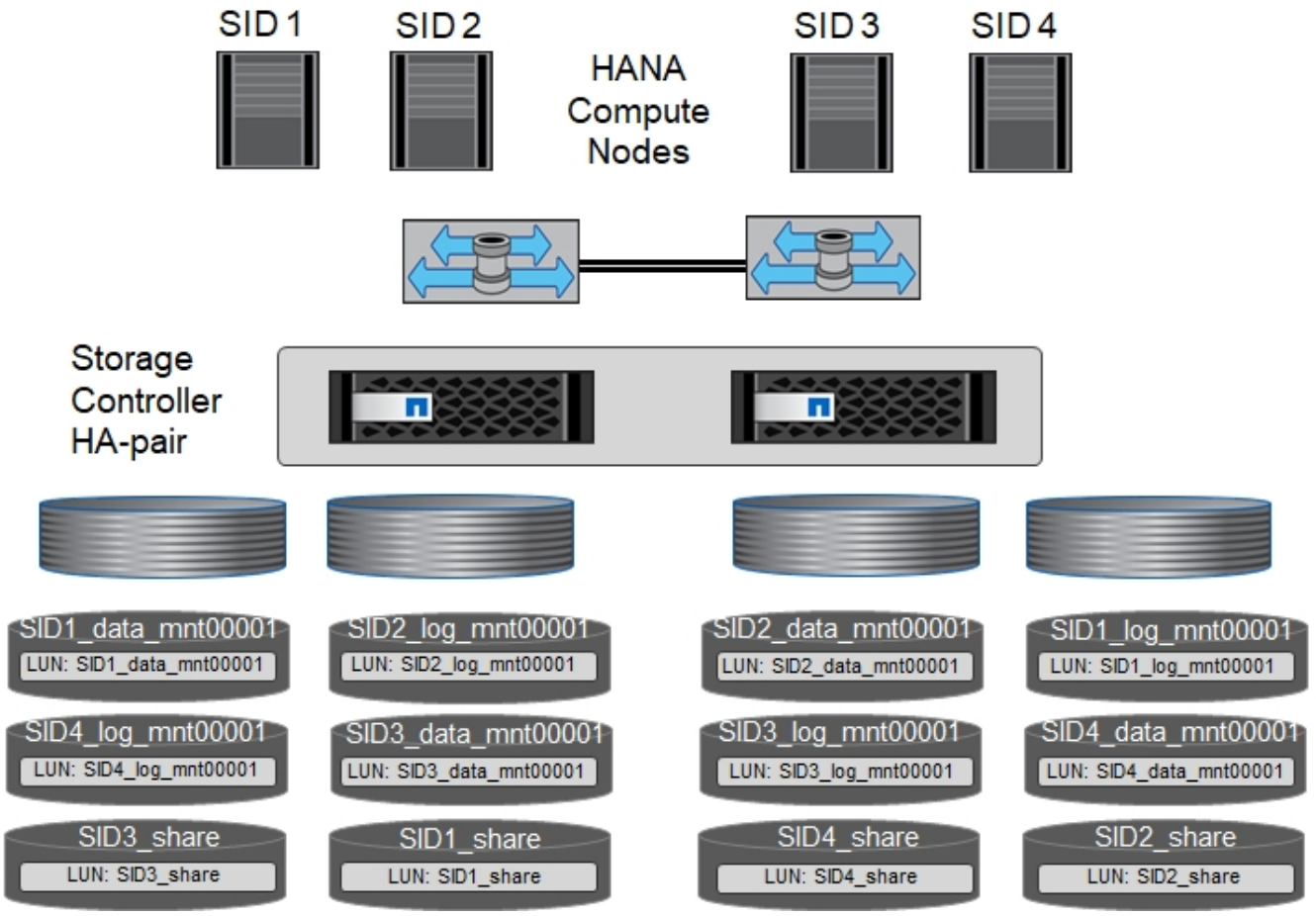
The following figure shows the list of initiators for SS3_HANA. The igroup contains all WWPNs of the servers and is assigned to the port set of the storage controller.




## Volume and LUN configuration for SAP HANA single-host systems

The following figure shows the volume configuration of four single-host SAP HANA systems. The data and log volumes of each SAP HANA system are distributed to different storage controllers. For example, volume `SID1``data``mnt00001 `is configured on controller A and volume `SID1` `log``mnt00001` is configured on controller B. Within each volume, a single LUN is configured.

> If only one storage controller of a high-availability (HA) pair is used for the SAP HANA systems, data volumes and log volumes can also be stored on the same storage controller.




For each SAP HANA host, a data volume, a log volume, and a volume for `/hana/shared` are configured. The following table shows an example configuration with four SAP HANA single-host systems.

| Purpose | Aggregate 1 at Controller A | Aggregate 2 at Controller A | Aggregate 1 at Controller B | Aggregate 2 at Controller B |
|---|---|---|---|---|
| Data, log, and shared volumes for system SID1 | Data volume: SID1_data_mnt00001 | Shared volume: SID1_shared | – | Log volume: SID1_log_mnt00001 |
| Data, log, and shared volumes for system SID2 | – | Log volume: SID2_log_mnt00001 | Data volume: SID2_data_mnt00001 | Shared volume: SID2_shared |
| Data, log, and shared volumes for system SID3 | Shared volume: SID3_shared | Data volume: SID3_data_mnt00001 | Log volume: SID3_log_mnt00001 | – |
| Data, log, and shared volumes for system SID4 | Log volume: SID4_log_mnt00001 | – | Shared volume: SID4_shared | Data volume: SID4_data_mnt00001 |

The next table shows an example of the mount point configuration for a single-host system.

| LUN | Mount point at HANA host | Note |
|---|---|---|
| SID1_data_mnt00001 | /hana/data/SID1/mnt00001 | Mounted using /etc/fstab entry |

| LUN | Mount point at HANA host | Note |
|---|---|---|
| SID1_log_mnt00001 | /hana/log/SID1/mnt00001 | Mounted using /etc/fstab entry |
| SID1_shared | /hana/shared/SID1 | Mounted using /etc/fstab entry |

> With the described configuration, the `/usr/sap/SID1` directory in which the default home directory of user SID1adm is stored, is on the local disk. In a disaster recovery setup with disk-based replication, NetApp recommends creating an additional LUN within the `SID1`_`shared `volume for the `/usr/sap/SID1` directory so that all file systems are on the central storage.

## Volume and LUN configuration for SAP HANA single-host systems using Linux LVM

The Linux LVM can be used to increase performance and to address LUN size limitations. The different LUNs of an LVM volume group should be stored within a different aggregate and at a different controller. The following table shows an example for two LUNs per volume group.

> It is not necessary to use LVM with multiple LUNs to fulfil the SAP HANA KPIs. A single LUN setup fulfils the required KPIs.

| Purpose | Aggregate 1 at Controller A | Aggregate 2 at Controller A | Aggregate 1 at Controller B | Aggregate 2 at Controller B |
|---|---|---|---|---|
| Data, log, and shared volumes for LVM based system | Data volume: SID1_data_mnt00001 | Shared volume: SID1_shared<br>Log2 volume: SID1_log2_mnt00001 | Data2 volume: SID1_data2_mnt00001 | Log volume: SID1_log_mnt00001 |

At the SAP HANA host, volume groups and logical volumes must be created and mounted. The next table lists the mount points for single-host systems using LVM.

| Logical volume/LUN | Mount point at SAP HANA host | Note |
|---|---|---|
| LV: SID1_data_mnt0000-vol | /hana/data/SID1/mnt00001 | Mounted using /etc/fstab entry |
| LV: SID1_log_mnt00001-vol | /hana/log/SID1/mnt00001 | Mounted using /etc/fstab entry |
| LUN: SID1_shared | /hana/shared/SID1 | Mounted using /etc/fstab entry |

> With the described configuration, the `/usr/sap/SID1` directory in which the default home directory of user SID1adm is stored, is on the local disk. In a disaster recovery setup with disk-based replication, NetApp recommends creating an additional LUN within the `SID1`_`shared `volume for the `/usr/sap/SID1` directory so that all file systems are on the central storage.

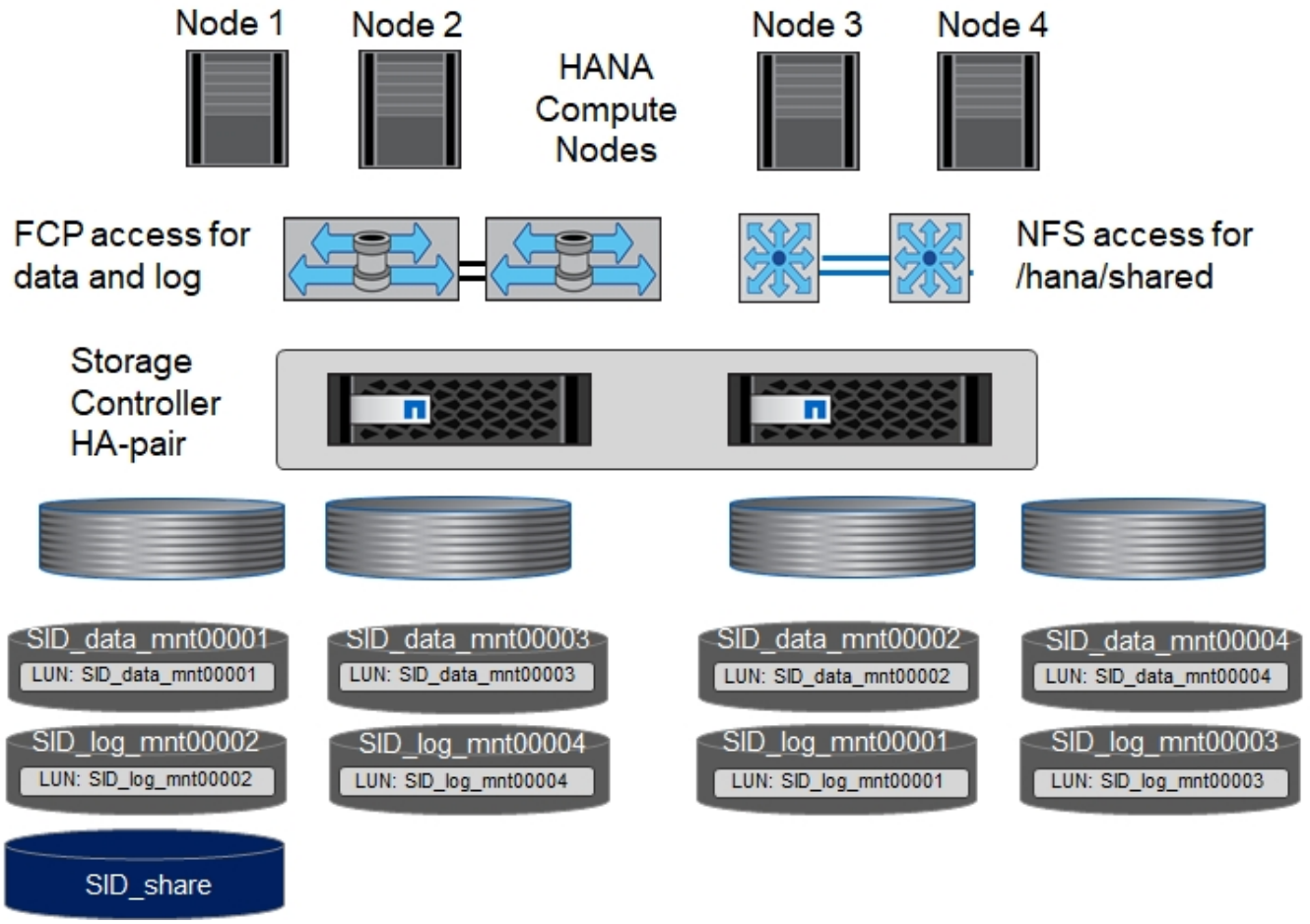
## Volume and LUN configuration for SAP HANA multiple-host systems

The following figure shows the volume configuration of a 4+1 multiple-host SAP HANA system. The data volumes and log volumes of each SAP HANA host are distributed to different storage controllers. For example, the volume `SID`` *data* ``mnt00001` is configured on controller A and the volume `SID`` *log* ``mnt00001` is

configured on controller B. One LUN is configured within each volume.

The `/hana/shared` volume must be accessible by all HANA hosts and is therefore exported by using NFS. Even though there are no specific performance KPIs for the `/hana/shared` file system, NetApp recommends using a 10Gb Ethernet connection.

> If only one storage controller of an HA pair is used for the SAP HANA system, data and log volumes can also be stored on the same storage controller.




For each SAP HANA host, a data volume and a log volume are created. The `/hana/shared` volume is used by all hosts of the SAP HANA system. The following figure shows an example configuration for a 4+1 multiple-host SAP HANA system.

| Purpose | Aggregate 1 at Controller A | Aggregate 2 at Controller A | Aggregate 1 at Controller B | Aggregate 2 at Controller B |
|---|---|---|---|---|
| Data and log volumes for node 1 | Data volume: SID_data_mnt00001 | – | Log volume: SID_log_mnt00001 | – |
| Data and log volumes for node 2 | Log volume: SID_log_mnt00002 | – | Data volume: SID_data_mnt00002 | – |
| Data and log volumes for node 3 | – | Data volume: SID_data_mnt00003 | – | Log volume: SID_log_mnt00003 |

| Purpose | Aggregate 1 at Controller A | Aggregate 2 at Controller A | Aggregate 1 at Controller B | Aggregate 2 at Controller B |
|---|---|---|---|---|
| Data and log volumes for node 4 | – | Log volume: SID_log_mnt00004 | – | Data volume: SID_data_mnt00004 |
| Shared volume for all hosts | Shared volume: SID_shared | – | – | – |

The next table shows the configuration and the mount points of a multiple-host system with four active SAP HANA hosts.

| LUN or Volume | Mount point at SAP HANA host | Note |
|---|---|---|
| LUN: SID_data_mnt00001 | /hana/data/SID/mnt00001 | Mounted using storage connector |
| LUN: SID_log_mnt00001 | /hana/log/SID/mnt00001 | Mounted using storage connector |
| LUN: SID_data_mnt00002 | /hana/data/SID/mnt00002 | Mounted using storage connector |
| LUN: SID_log_mnt00002 | /hana/log/SID/mnt00002 | Mounted using storage connector |
| LUN: SID_data_mnt00003 | /hana/data/SID/mnt00003 | Mounted using storage connector |
| LUN: SID_log_mnt00003 | /hana/log/SID/mnt00003 | Mounted using storage connector |
| LUN: SID_data_mnt00004 | /hana/data/SID/mnt00004 | Mounted using storage connector |
| LUN: SID_log_mnt00004 | /hana/log/SID/mnt00004 | Mounted using storage connector |
| Volume: SID_shared | /hana/shared/SID | Mounted at all hosts using NFS and /etc/fstab entry |

With the described configuration, the `/usr/sap/SID` directory in which the default home directory of user SIDadm is stored is on the local disk for each HANA host. In a disaster recovery setup with disk-based replication, NetApp recommends creating four additional subdirectories in the `SID`_`shared` volume for the `/usr/sap/SID` file system so that each database host has all its file systems on the central storage.

## Volume and LUN configuration for SAP HANA multiple-host systems using Linux LVM

The Linux LVM can be used to increase performance and to address LUN size limitations. The different LUNs of an LVM volume group should be stored within a different aggregate and at a different controller. The following table shows an example for two LUNs per volume group for a 2+1 SAP HANA multiple host system.

It is not necessary to use LVM to combine several LUN to fulfil the SAP HANA KPIs. A single LUN setup fulfils the required KPIs.

| Purpose | Aggregate 1 at Controller A | Aggregate 2 at Controller A | Aggregate 1 at Controller B | Aggregate 2 at Controller B |
|---|---|---|---|---|
| Data and log volumes for node 1 | Data volume: SID_data_mnt00001 | Log2 volume: SID_log2_mnt00001 | Log volume: SID_log_mnt00001 | Data2 volume: SID_data2_mnt00001 |

| Purpose | Aggregate 1 at Controller A | Aggregate 2 at Controller A | Aggregate 1 at Controller B | Aggregate 2 at Controller B |
|---|---|---|---|---|
| Data and log volumes for node 2 | Log2 volume: SID_log2_mnt00002 | Data volume: SID_data_mnt00002 | Data2 volume: SID_data2_mnt00002 | Log volume: SID_log_mnt00002 |
| Shared volume for all hosts | Shared volume: SID_shared | – | – | – |

At the SAP HANA host, volume groups and logical volumes need to be created and mounted:

| Logical volume (LV) or volume | Mount point at SAP HANA host | Note |
|---|---|---|
| LV: SID_data_mnt00001-vol | /hana/data/SID/mnt00001 | Mounted using storage connector |
| LV: SID_log_mnt00001-vol | /hana/log/SID/mnt00001 | Mounted using storage connector |
| LV: SID_data_mnt00002-vol | /hana/data/SID/mnt00002 | Mounted using storage connector |
| LV: SID_log_mnt00002-vol | /hana/log/SID/mnt00002 | Mounted using storage connector |
| Volume: SID_shared | /hana/shared | Mounted at all hosts using NFS and /etc/fstab entry |

> With the described configuration, the `/usr/sap/SID` directory in which the default home directory of user SIDadm is stored, is on the local disk for each HANA host. In a disaster recovery setup with disk-based replication, NetApp recommends creating four additional subdirectories in the `SID`_`shared` volume for the `/usr/sap/SID` file system so that each database host has all its file systems on the central storage.

## Volume options

The volume options listed in the following table must be verified and set on all SVMs.

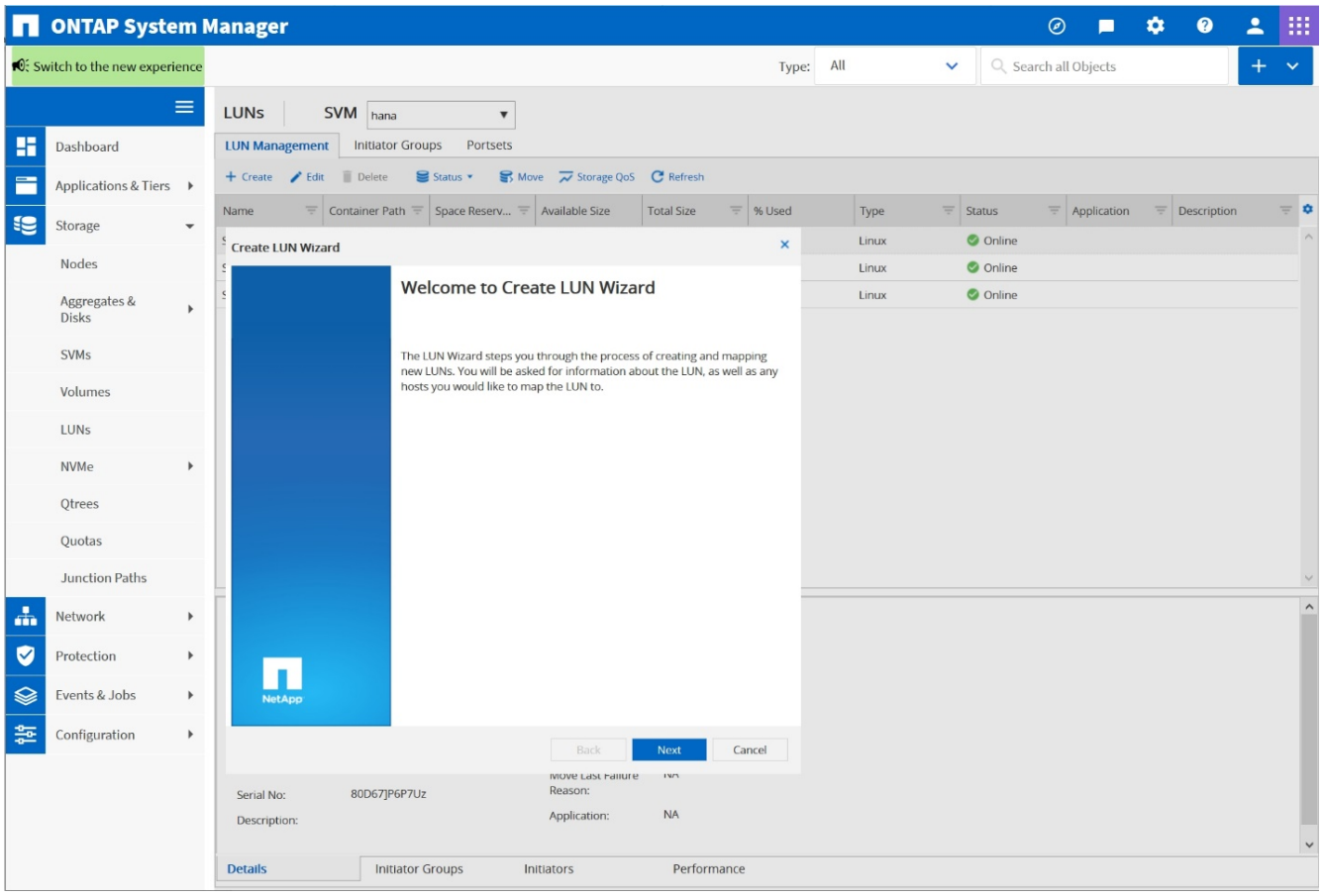
| Action | ONTAP 9 |
|---|---|
| Disable automatic Snapshot copies | vol modify –vserver <vserver-name> -volume <volname> -snapshot-policy none |
| Disable visibility of Snapshot directory | vol modify -vserver <vserver-name> -volume <volname> -snapdir-access false |

## Creating LUNs, volumes, and mapping LUNs to initiator groups

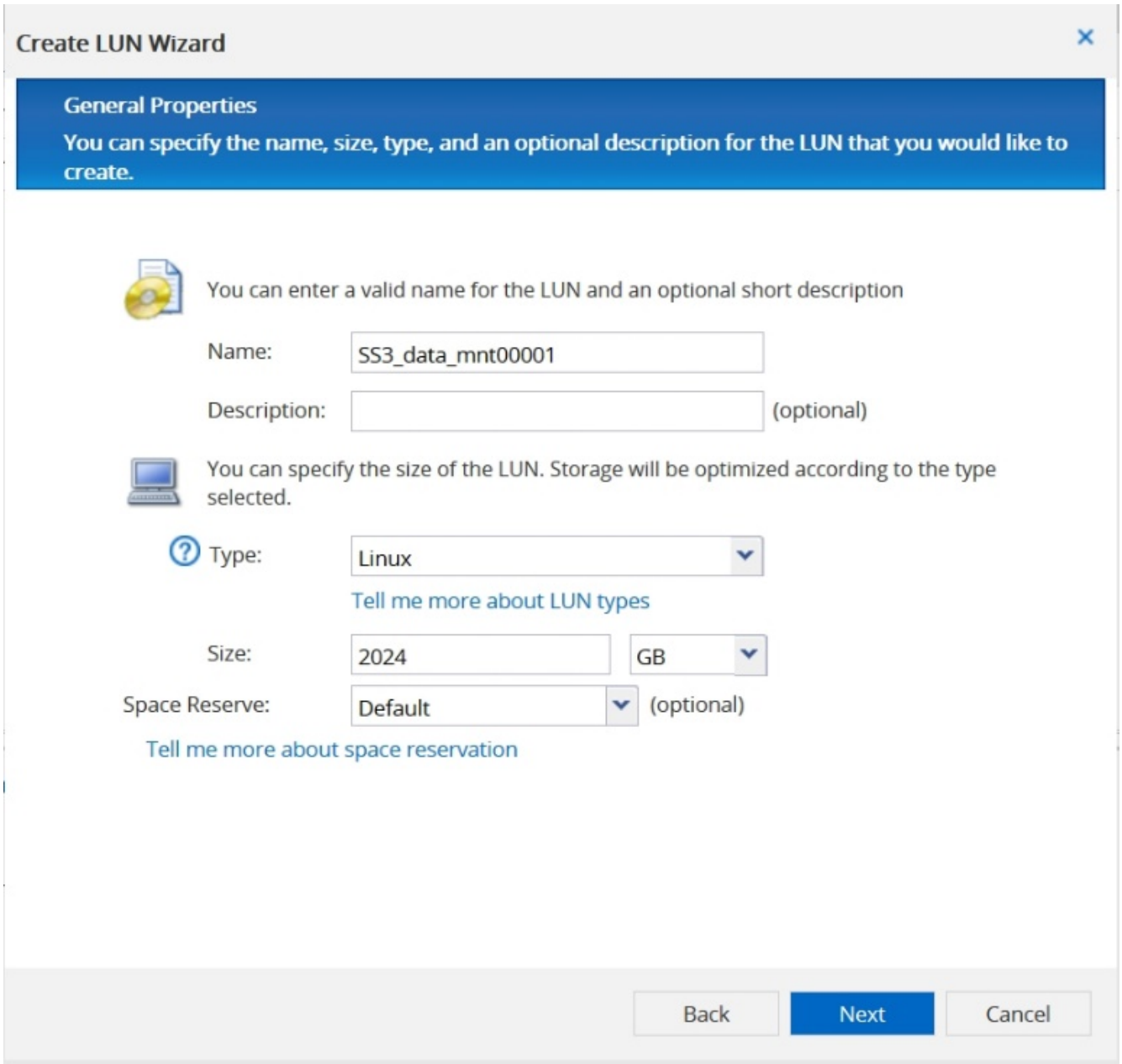
You can use NetApp OnCommand System Manager to create storage volumes and LUNs and the map them to the igroups of the servers.

The following steps show the configuration of a 2+1 multiple-host HANA system with the SID SS3.

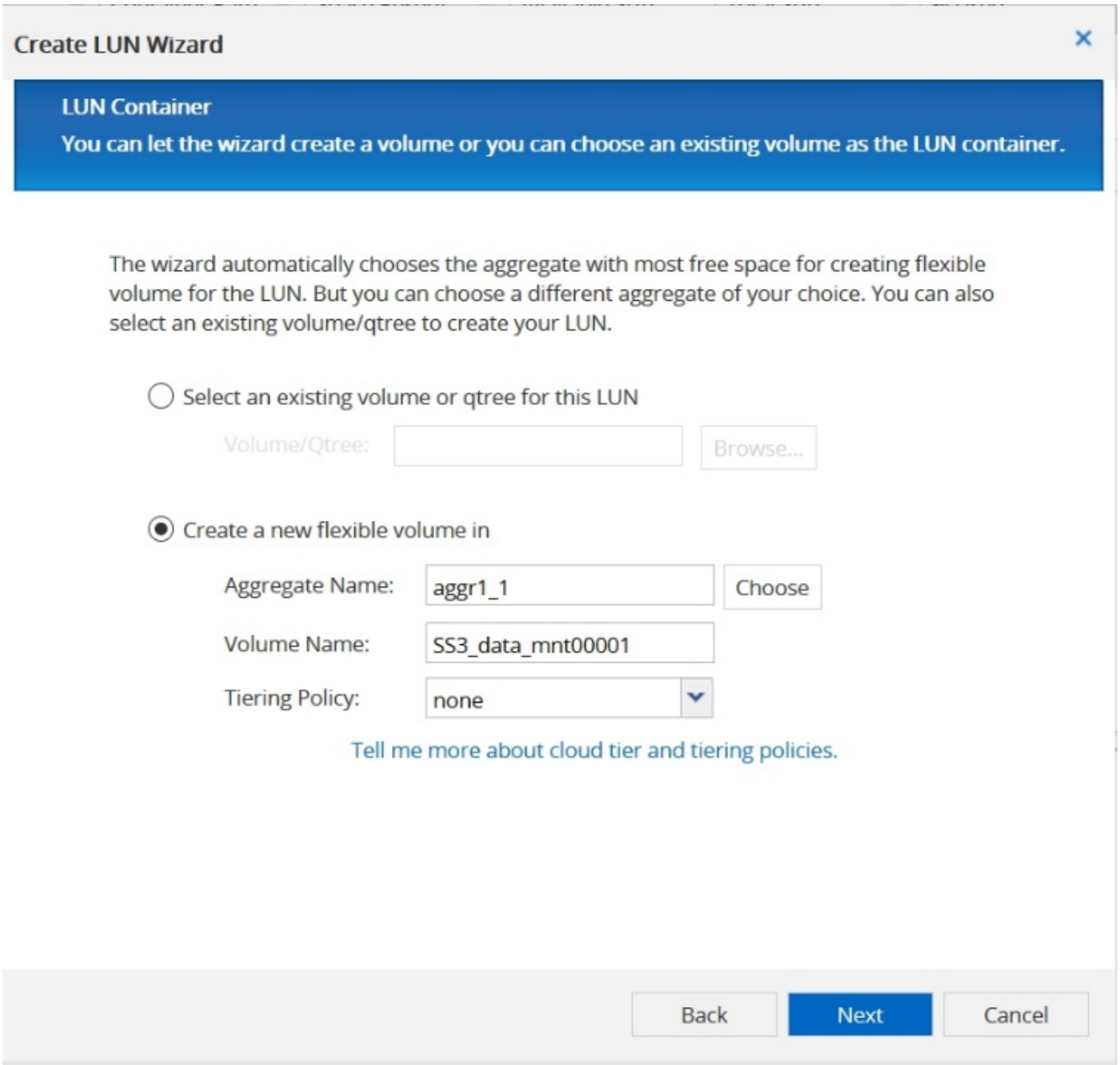
1. Start the Create LUN Wizard in NetApp ONTAP System Manager.



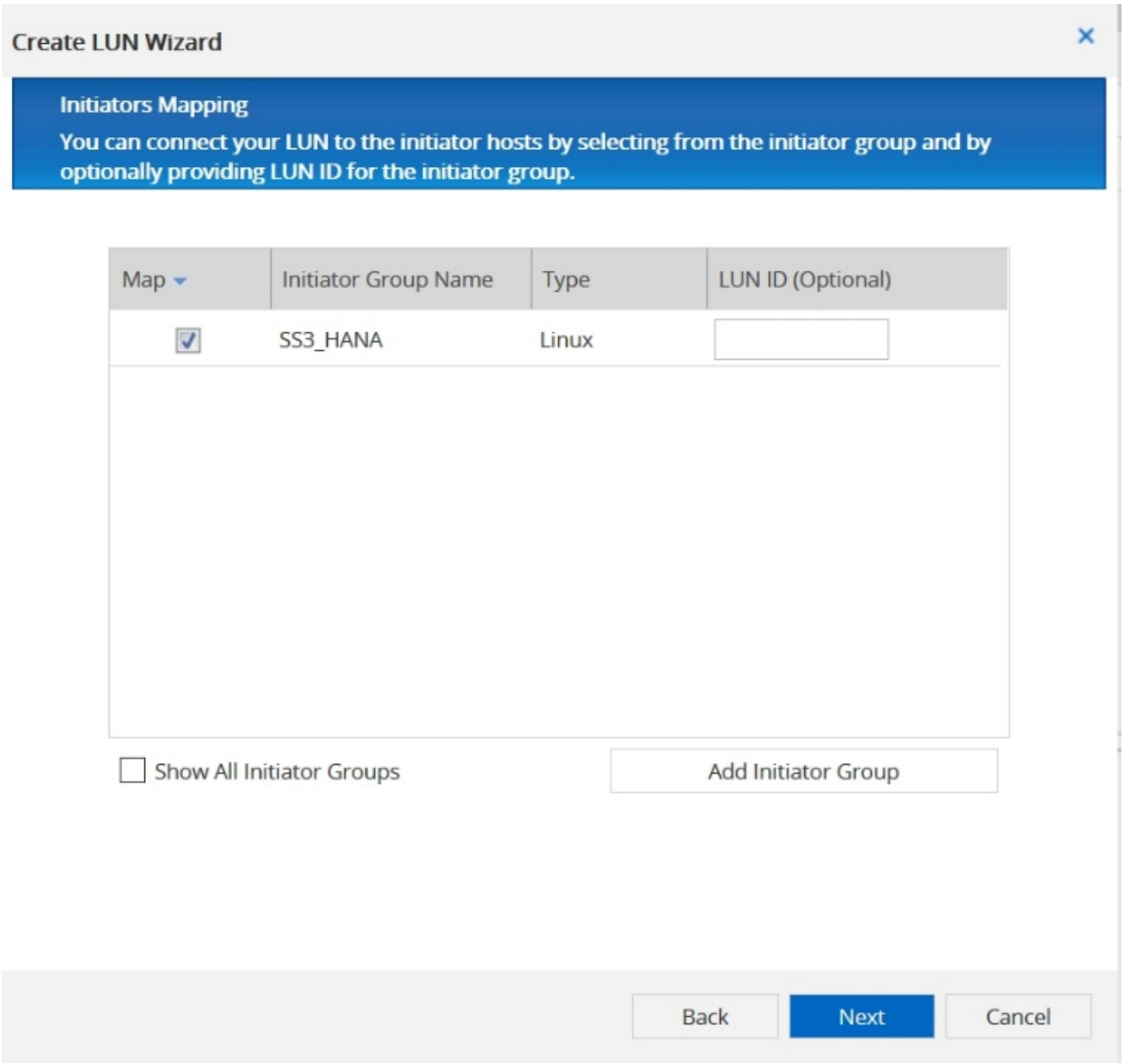

2. Enter the LUN name, select the LUN type, and enter the size of the LUN.



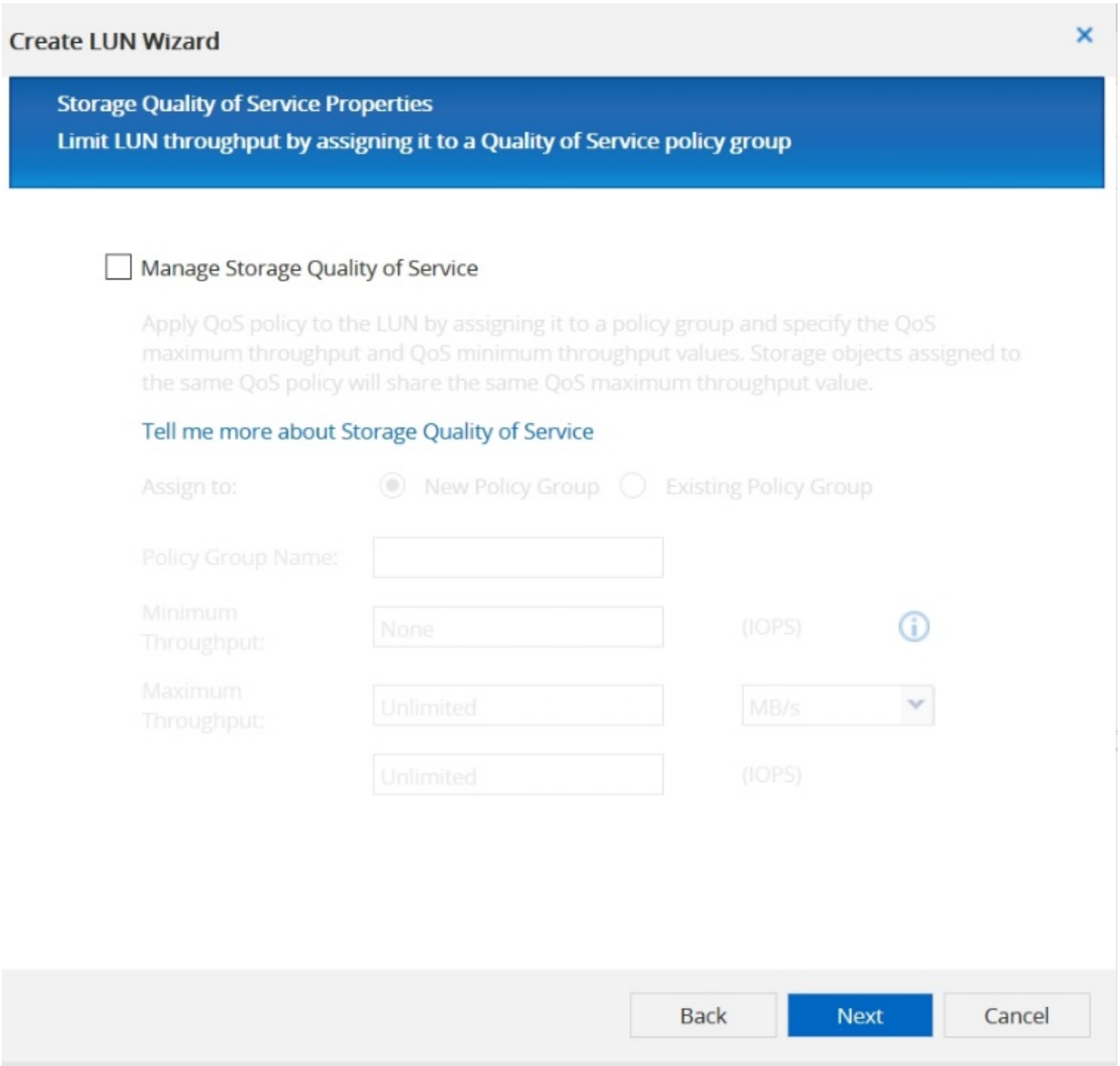

3. Enter the volume name and the hosting aggregate.



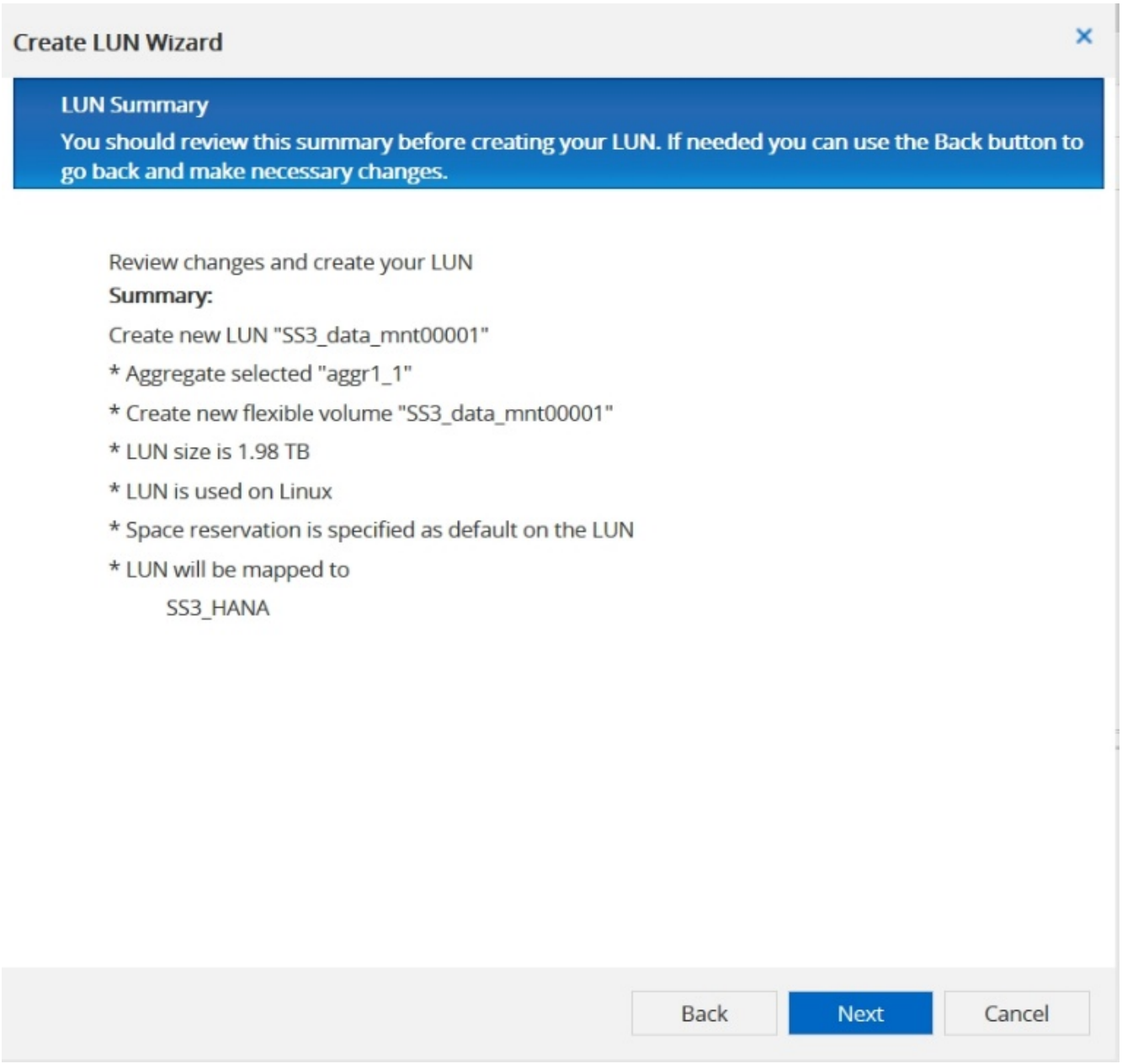

4. Select the igroups to which the LUNs should be mapped.



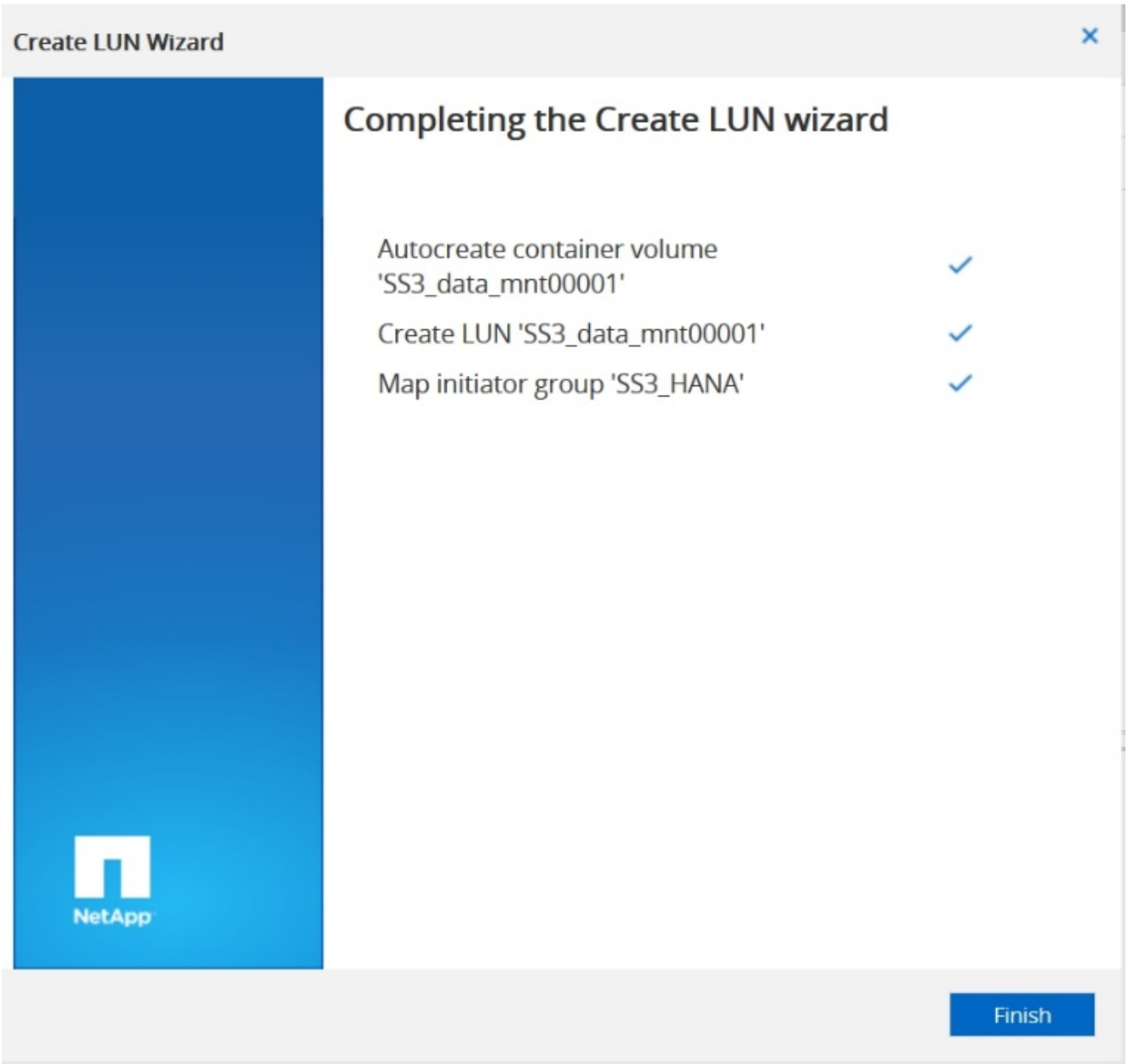

5. Provide the QoS settings.




6. Click Next on the Summary page.

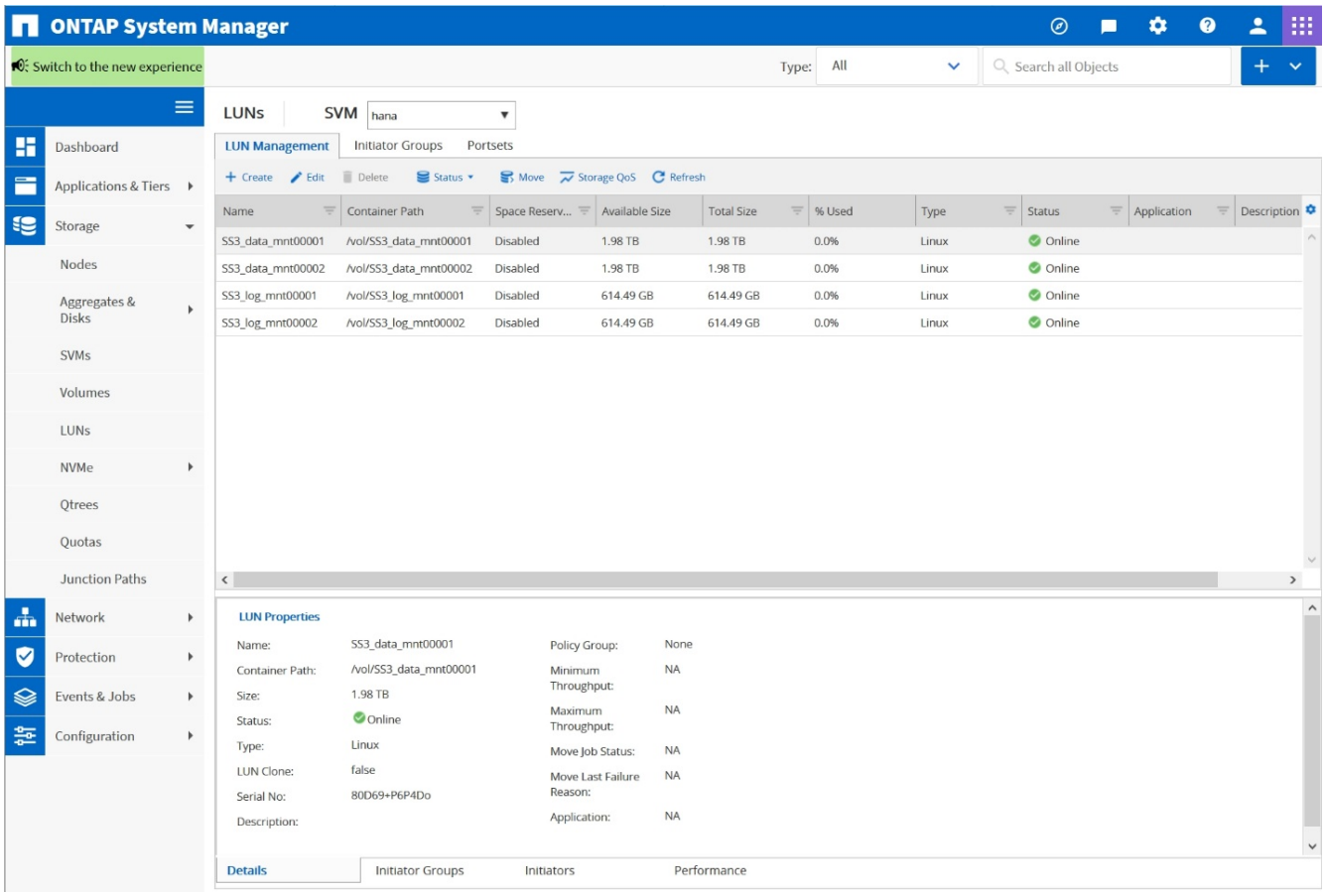



7. Click Finish on the Completion page.




8. Repeat steps 2 to 7 for each LUN.

   The following figure shows a summary of all LUNs that need to be created for 2+1 multiple-host setup.




## Creating LUNs, volumes, and mapping LUNs to igroups using the CLI

This section shows an example configuration using the command line with ONTAP 9.8 for a 2+1 SAP HANA multiple host system with SID FC5 using LVM and two LUNs per LVM volume group.

1. Create all necessary volumes.

```
vol create -volume FC5_data_mnt00001 -aggregate aggr1_1 -size 1200g
-snapshot-policy none -foreground true -encrypt false  -space-guarantee
none
vol create -volume FC5_log_mnt00002  -aggregate aggr2_1 -size 280g
-snapshot-policy none -foreground true -encrypt false  -space-guarantee
none
vol create -volume FC5_log_mnt00001  -aggregate aggr1_2 -size 280g
-snapshot-policy none -foreground true -encrypt false -space-guarantee
none
vol create -volume FC5_data_mnt00002  -aggregate aggr2_2 -size 1200g
-snapshot-policy none -foreground true -encrypt false -space-guarantee
none
vol create -volume FC5_data2_mnt00001 -aggregate aggr1_2 -size 1200g
-snapshot-policy none -foreground true -encrypt false -space-guarantee
none
vol create -volume FC5_log2_mnt00002  -aggregate aggr2_2 -size 280g
-snapshot-policy none -foreground true -encrypt false -space-guarantee
none
vol create -volume FC5_log2_mnt00001  -aggregate aggr1_1 -size 280g
-snapshot-policy none -foreground true -encrypt false  -space-guarantee
none
vol create -volume FC5_data2_mnt00002  -aggregate aggr2_1 -size 1200g
-snapshot-policy none -foreground true -encrypt false -space-guarantee
none
vol create -volume FC5_shared -aggregate aggr1_1 -size 512g -state
online -policy default -snapshot-policy none -junction-path /FC5_shared
-encrypt false  -space-guarantee none
```

2. Create all LUNs.

```
lun create -path  /vol/FC5_data_mnt00001/FC5_data_mnt00001   -size 1t
-ostype linux -space-reserve disabled -space-allocation disabled -class
regular
lun create -path /vol/FC5_data2_mnt00001/FC5_data2_mnt00001 -size 1t
-ostype linux -space-reserve disabled -space-allocation disabled -class
regular
lun create -path /vol/FC5_data_mnt00002/FC5_data_mnt00002 -size 1t
-ostype linux -space-reserve disabled -space-allocation disabled -class
regular
lun create -path /vol/FC5_data2_mnt00002/FC5_data2_mnt00002 -size 1t
-ostype linux -space-reserve disabled -space-allocation disabled -class
regular
lun create -path /vol/FC5_log_mnt00001/FC5_log_mnt00001 -size 260g
-ostype linux -space-reserve disabled -space-allocation disabled -class
regular
lun create -path /vol/FC5_log2_mnt00001/FC5_log2_mnt00001 -size 260g
-ostype linux -space-reserve disabled -space-allocation disabled -class
regular
lun create -path /vol/FC5_log_mnt00002/FC5_log_mnt00002 -size 260g
-ostype linux -space-reserve disabled -space-allocation disabled -class
regular
lun create -path /vol/FC5_log2_mnt00002/FC5_log2_mnt00002 -size 260g
-ostype linux -space-reserve disabled -space-allocation disabled -class
regular
```

3. Create the igroup for all servers belonging to system FC5.

```
lun igroup create -igroup HANA-FC5 -protocol fcp -ostype linux
-initiator 10000090fadcc5fa,10000090fadcc5fb,
10000090fadcc5c1,10000090fadcc5c2,  10000090fadcc5c3,10000090fadcc5c4
-vserver hana
```

4. Map all LUNs to the created igroup.

```
lun map -path  /vol/FC5_data_mnt00001/FC5_data_mnt00001    -igroup HANA-
FC5
lun map -path /vol/FC5_data2_mnt00001/FC5_data2_mnt00001  -igroup HANA-
FC5
lun map -path /vol/FC5_data_mnt00002/FC5_data_mnt00002  -igroup HANA-FC5
lun map -path /vol/FC5_data2_mnt00002/FC5_data2_mnt00002  -igroup HANA-
FC5
lun map -path /vol/FC5_log_mnt00001/FC5_log_mnt00001  -igroup HANA-FC5
lun map -path /vol/FC5_log2_mnt00001/FC5_log2_mnt00001  -igroup HANA-FC5
lun map -path /vol/FC5_log_mnt00002/FC5_log_mnt00002  -igroup HANA-FC5
lun map -path /vol/FC5_log2_mnt00002/FC5_log2_mnt00002  -igroup HANA-FC5
```

Next: SAP HANA storage connector API.

# SAP HANA storage connector API

Previous: Storage controller setup.

A storage connector is required only in multiple-host environments that have failover capabilities. In multiple-host setups, SAP HANA provides high-availability functionality so that an SAP HANA database host can fail over to a standby host. In this case, the LUNs of the failed host are accessed and used by the standby host. The storage connector is used to make sure that a storage partition can be actively accessed by only one database host at a time.

In SAP HANA multiple-host configurations with NetApp storage, the standard storage connector delivered by SAP is used. The “SAP HANA FC Storage Connector Admin Guide” can be found as an attachment to SAP note 1900823.

Next: Host setup.

# Host setup

Previous: SAP HANA storage connector API.

Before setting up the host, NetApp SAN Host Utilities must be downloaded from the NetApp Support site and installed on the HANA servers. The Host Utility documentation includes information about additional software that must be installed depending on the FCP HBA used.

The documentation also contains information about multipath configurations that are specific to the Linux version used. This document covers the required configuration steps for SLES 15 and Red Hat Enterprise Linux 7.6 or higher, as described in the Linux Host Utilities 7.1 Installation and Setup Guide.

## Configure multipathing

> Steps 1 to 6 must be performed on all worker and standby hosts in the SAP HANA multiple-host configuration.

To configure multipathing, complete the following steps:

1. Run the Linux `rescan-scsi-bus.sh -a` command on each server to discover new LUNs.
2. Run the `sanlun lun show` command and verify that all required LUNs are visible. The following example shows the `sanlun lun show` command output for a 2+1 multiple-host HANA system with two data LUNs and two log LUNs. The output shows the LUNs and the corresponding device files, such as LUN `SS3_data_mnt00001` and the device file `/dev/sdag`. Each LUN has eight FC paths from the host to the storage controllers.

```
stlrx300s8-6:~ # sanlun lun show
controller(7mode/E-Series)/
device          host       lun
vserver(cDOT/FlashRay)        lun-pathname
filename        adapter    protocol   size    product
------------------------------------------------------------------------
-------------------------------------------------
hana                          /vol/SS3_log_mnt00002/SS3_log_mnt00002
/dev/sdah       host11     FCP        512.0g  cDOT
hana                          /vol/SS3_data_mnt00001/SS3_data_mnt00001
/dev/sdag       host11     FCP        1.2t    cDOT
hana                          /vol/SS3_data_mnt00002/SS3_data_mnt00002
/dev/sdaf       host11     FCP        1.2t    cDOT
hana                          /vol/SS3_log_mnt00002/SS3_log_mnt00002
/dev/sdae       host11     FCP        512.0g  cDOT
hana                          /vol/SS3_data_mnt00001/SS3_data_mnt00001
/dev/sdad       host11     FCP        1.2t    cDOT
hana                          /vol/SS3_data_mnt00002/SS3_data_mnt00002
/dev/sdac       host11     FCP        1.2t    cDOT
hana                          /vol/SS3_log_mnt00002/SS3_log_mnt00002
/dev/sdab       host11     FCP        512.0g  cDOT
hana                          /vol/SS3_data_mnt00001/SS3_data_mnt00001
/dev/sdaa       host11     FCP        1.2t    cDOT
hana                          /vol/SS3_data_mnt00002/SS3_data_mnt00002
/dev/sdz        host11     FCP        1.2t    cDOT
hana                          /vol/SS3_log_mnt00002/SS3_log_mnt00002
/dev/sdy        host11     FCP        512.0g  cDOT
hana                          /vol/SS3_data_mnt00001/SS3_data_mnt00001
/dev/sdx        host11     FCP        1.2t    cDOT
hana                          /vol/SS3_data_mnt00002/SS3_data_mnt00002
/dev/sdw        host11     FCP        1.2t    cDOT
hana                          /vol/SS3_log_mnt00001/SS3_log_mnt00001
/dev/sdv        host11     FCP        512.0g  cDOT
hana                          /vol/SS3_log_mnt00001/SS3_log_mnt00001
/dev/sdu        host11     FCP        512.0g  cDOT
hana                          /vol/SS3_log_mnt00001/SS3_log_mnt00001
/dev/sdt        host11     FCP        512.0g  cDOT
hana                          /vol/SS3_log_mnt00001/SS3_log_mnt00001
/dev/sds        host11     FCP        512.0g  cDOT
```

```
hana                            /vol/SS3_log_mnt00002/SS3_log_mnt00002
/dev/sdr        host10     FCP        512.0g  cDOT
hana                            /vol/SS3_data_mnt00001/SS3_data_mnt00001
/dev/sdq        host10     FCP        1.2t    cDOT
hana                            /vol/SS3_data_mnt00002/SS3_data_mnt00002
/dev/sdp        host10     FCP        1.2t    cDOT
hana                            /vol/SS3_log_mnt00002/SS3_log_mnt00002
/dev/sdo        host10     FCP        512.0g  cDOT
hana                            /vol/SS3_data_mnt00001/SS3_data_mnt00001
/dev/sdn        host10     FCP        1.2t    cDOT
hana                            /vol/SS3_data_mnt00002/SS3_data_mnt00002
/dev/sdm        host10     FCP        1.2t    cDOT
hana                            /vol/SS3_log_mnt00002/SS3_log_mnt00002
/dev/sdl        host10     FCP        512.0g  cDOT
hana                            /vol/SS3_data_mnt00001/SS3_data_mnt00001
/dev/sdk        host10     FCP        1.2t    cDOT
hana                            /vol/SS3_data_mnt00002/SS3_data_mnt00002
/dev/sdj        host10     FCP        1.2t    cDOT
hana                            /vol/SS3_log_mnt00002/SS3_log_mnt00002
/dev/sdi        host10     FCP        512.0g  cDOT
hana                            /vol/SS3_data_mnt00001/SS3_data_mnt00001
/dev/sdh        host10     FCP        1.2t    cDOT
hana                            /vol/SS3_data_mnt00002/SS3_data_mnt00002
/dev/sdg        host10     FCP        1.2t    cDOT
hana                            /vol/SS3_log_mnt00001/SS3_log_mnt00001
/dev/sdf        host10     FCP        512.0g  cDOT
hana                            /vol/SS3_log_mnt00001/SS3_log_mnt00001
/dev/sde        host10     FCP        512.0g  cDOT
hana                            /vol/SS3_log_mnt00001/SS3_log_mnt00001
/dev/sdd        host10     FCP        512.0g  cDOT
hana                            /vol/SS3_log_mnt00001/SS3_log_mnt00001
/dev/sdc        host10     FCP        512.0g  cDOT
```

3. Run the `multipath -r` command to get the worldwide identifiers (WWIDs) for the device file names:

> In this example, there are four LUNs.

```
stlrx300s8-6:~ # multipath -r
create: 3600a098038304436375d4d442d753878 undef NETAPP,LUN C-Mode
size=512G features='3 pg_init_retries 50 queue_if_no_path' hwhandler='0'
wp=undef
|-+- policy='service-time 0' prio=50 status=undef
| |- 10:0:1:0 sdd  8:48   undef ready running
| |- 10:0:3:0 sdf  8:80   undef ready running
| |- 11:0:0:0 sds  65:32  undef ready running
```

```
| `- 11:0:2:0 sdu  65:64  undef ready running
`-+- policy='service-time 0' prio=10 status=undef
  |- 10:0:0:0 sdc  8:32   undef ready running
  |- 10:0:2:0 sde  8:64   undef ready running
  |- 11:0:1:0 sdt  65:48  undef ready running
  `- 11:0:3:0 sdv  65:80  undef ready running
create: 3600a098038304436375d4d442d753879 undef NETAPP,LUN C-Mode
size=1.2T features='3 pg_init_retries 50 queue_if_no_path' hwhandler='0'
wp=undef
|-+- policy='service-time 0' prio=50 status=undef
| |- 10:0:1:1 sdj  8:144  undef ready running
| |- 10:0:3:1 sdp  8:240  undef ready running
| |- 11:0:0:1 sdw  65:96  undef ready running
| `- 11:0:2:1 sdac 65:192 undef ready running
`-+- policy='service-time 0' prio=10 status=undef
  |- 10:0:0:1 sdg  8:96   undef ready running
  |- 10:0:2:1 sdm  8:192  undef ready running
  |- 11:0:1:1 sdz  65:144 undef ready running
  `- 11:0:3:1 sdaf 65:240 undef ready running
create: 3600a098038304436392b4d442d6f534f undef NETAPP,LUN C-Mode
size=1.2T features='3 pg_init_retries 50 queue_if_no_path' hwhandler='0'
wp=undef
|-+- policy='service-time 0' prio=50 status=undef
| |- 10:0:0:2 sdh  8:112  undef ready running
| |- 10:0:2:2 sdn  8:208  undef ready running
| |- 11:0:1:2 sdaa 65:160 undef ready running
| `- 11:0:3:2 sdag 66:0   undef ready running
`-+- policy='service-time 0' prio=10 status=undef
  |- 10:0:1:2 sdk  8:160  undef ready running
  |- 10:0:3:2 sdq  65:0   undef ready running
  |- 11:0:0:2 sdx  65:112 undef ready running
  `- 11:0:2:2 sdad 65:208 undef ready running
create: 3600a098038304436392b4d442d6f5350 undef NETAPP,LUN C-Mode
size=512G features='3 pg_init_retries 50 queue_if_no_path' hwhandler='0'
wp=undef
|-+- policy='service-time 0' prio=50 status=undef
| |- 10:0:0:3 sdi  8:128  undef ready running
| |- 10:0:2:3 sdo  8:224  undef ready running
| |- 11:0:1:3 sdab 65:176 undef ready running
| `- 11:0:3:3 sdah 66:16  undef ready running
`-+- policy='service-time 0' prio=10 status=undef
  |- 10:0:1:3 sdl  8:176  undef ready running
  |- 10:0:3:3 sdr  65:16  undef ready running
  |- 11:0:0:3 sdy  65:128 undef ready running
  `- 11:0:2:3 sdae 65:224 undef ready running
```

4. Edit the `/etc/multipath.conf` file and add the WWIDs and alias names.

> The example output shows the content of the `/etc/multipath.conf` file, which includes alias names for the four LUNs of a 2+1 multiple-host system. If there is no `multipath.conf` file available, you can create one by running the following command: `multipath -T > /etc/multipath.conf`.

```
stlrx300s8-6:/ # cat /etc/multipath.conf
multipaths {
        multipath {
                wwid    3600a098038304436392b4d442d6f534f
                alias   hana-SS3_data_mnt00001
        }
        multipath {
                wwid    3600a098038304436375d4d442d753879
                alias   hana-SS3_data_mnt00002
        }
        multipath {
                wwid    3600a098038304436375d4d442d753878
                alias   hana-SS3_log_mnt00001
        }
        multipath {
                wwid    3600a098038304436392b4d442d6f5350
                alias   hana-SS3_log_mnt00002
        }
}
```

5. Run the `multipath -r` command to reload the device map.
6. Verify the configuration by running the `multipath -ll` command to list all the LUNs, alias names, and active and standby paths.

> The following example output shows the output of a 2+1 multiple-host HANA system with two data and two log LUNs.

```
stlrx300s8-6:~ # multipath -ll
hana- SS3_data_mnt00002 (3600a098038304436375d4d442d753879) dm-1
NETAPP,LUN C-Mode
size=1.2T features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handler' hwhandler='1 alua' wp=rw
|-+- policy='service-time 0' prio=50 status=enabled
| |- 10:0:1:1 sdj  8:144  active ready running
| |- 10:0:3:1 sdp  8:240  active ready running
| |- 11:0:0:1 sdw  65:96  active ready running
| `- 11:0:2:1 sdac 65:192 active ready running
`-+- policy='service-time 0' prio=10 status=enabled
```

```
  |- 10:0:0:1 sdg  8:96   active ready running
  |- 10:0:2:1 sdm  8:192  active ready running
  |- 11:0:1:1 sdz  65:144 active ready running
  `- 11:0:3:1 sdaf 65:240 active ready running
hana- SS3_data_mnt00001 (3600a098038304436392b4d442d6f534f) dm-2
NETAPP,LUN C-Mode
size=1.2T features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handler' hwhandler='1 alua' wp=rw
|-+- policy='service-time 0' prio=50 status=enabled
| |- 10:0:0:2 sdh  8:112  active ready running
| |- 10:0:2:2 sdn  8:208  active ready running
| |- 11:0:1:2 sdaa 65:160 active ready running
| `- 11:0:3:2 sdag 66:0   active ready running
`-+- policy='service-time 0' prio=10 status=enabled
  |- 10:0:1:2 sdk  8:160  active ready running
  |- 10:0:3:2 sdq  65:0   active ready running
  |- 11:0:0:2 sdx  65:112 active ready running
  `- 11:0:2:2 sdad 65:208 active ready running
hana- SS3_log_mnt00002 (3600a098038304436392b4d442d6f5350) dm-3
NETAPP,LUN C-Mode
size=512G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handler' hwhandler='1 alua' wp=rw
|-+- policy='service-time 0' prio=50 status=enabled
| |- 10:0:0:3 sdi  8:128  active ready running
| |- 10:0:2:3 sdo  8:224  active ready running
| |- 11:0:1:3 sdab 65:176 active ready running
| `- 11:0:3:3 sdah 66:16  active ready running
`-+- policy='service-time 0' prio=10 status=enabled
  |- 10:0:1:3 sdl  8:176  active ready running
  |- 10:0:3:3 sdr  65:16  active ready running
  |- 11:0:0:3 sdy  65:128 active ready running
  `- 11:0:2:3 sdae 65:224 active ready running
hana- SS3_log_mnt00001 (3600a098038304436375d4d442d753878) dm-0
NETAPP,LUN C-Mode
size=512G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handler' hwhandler='1 alua' wp=rw
|-+- policy='service-time 0' prio=50 status=enabled
| |- 10:0:1:0 sdd  8:48   active ready running
| |- 10:0:3:0 sdf  8:80   active ready running
| |- 11:0:0:0 sds  65:32  active ready running
| `- 11:0:2:0 sdu  65:64  active ready running
`-+- policy='service-time 0' prio=10 status=enabled
  |- 10:0:0:0 sdc  8:32   active ready running
  |- 10:0:2:0 sde  8:64   active ready running
  |- 11:0:1:0 sdt  65:48  active ready running
  `- 11:0:3:0 sdv  65:80  active ready running
```

## Create LVM volume groups and logical volumes

This step is only needed if LVM will be used. The following example is for a 2+1 host setup using SID FC5.

> For an LVM-based setup, the multipath configuration described in the previous section must be completed as well. In this example, eight LUNs must be configured for multipathing.

1. Initialize all LUNs as a physical volume.

```
pvcreate /dev/mapper/hana-FC5_data_mnt00001
pvcreate /dev/mapper/hana-FC5_data2_mnt00001pvcreate /dev/mapper/hana-
FC5_data_mnt00002
pvcreate /dev/mapper/hana-FC5_data2_mnt00002
pvcreate /dev/mapper/hana-FC5_log_mnt00001
pvcreate /dev/mapper/hana-FC5_log2_mnt00001pvcreate /dev/mapper/hana-
FC5_log_mnt00002
pvcreate /dev/mapper/hana-FC5_log2_mnt00002
```

2. Create the volume groups for each data and log partition.

```
vgcreate FC5_data_mnt00001 /dev/mapper/hana-FC5_data_mnt00001
/dev/mapper/hana-FC5_data2_mnt00001
vgcreate FC5_data_mnt00002 /dev/mapper/hana-FC5_data_mnt00002
/dev/mapper/hana-FC5_data2_mnt00002
vgcreate FC5_log_mnt00001 /dev/mapper/hana-FC5_log_mnt00001
/dev/mapper/hana-FC5_log2_mnt00001
vgcreate FC5_log_mnt00002 /dev/mapper/hana-FC5_log_mnt00002
/dev/mapper/hana-FC5_log2_mnt00002
```

3. Create a logical volume for each data and log partition. Use a stripe size that is equal to the number of LUNs used per volume group (in example two) and a stripe size of 256k for data and 64k for log. SAP only supports one logical volume per volume group.

```
lvcreate --extents 100%FREE -i 2 -I 256k --name vol FC5_data_mnt00001
lvcreate --extents 100%FREE -i 2 -I 256k --name vol FC5_data_mnt00002
lvcreate --extents 100%FREE -i 2 -I 64k --name vol FC5_log_mnt00002
lvcreate --extents 100%FREE -i 2 -I 64k --name vol FC5_log_mnt00001
```

4. Scan the physical volumes, volume groups, and vol groups at all other hosts.

```
modprobe dm_mod
pvscan
vgscan
lvscan
```

If the above commands do not find the volumes, a restart is required.

5. To mount the logical volumes, the logical volumes must be activated. To activate the volumes, run the following command:

```
vgchange -a y
```

## Create file systems

To create the XFS file system on each LUN belonging to the HANA system, take one of the following actions:

- For a single-host system, create the XFS file system on the data, log, and `/hana/shared` LUNs.

```
stlrx300s8-6:/ # mkfs.xfs /dev/mapper/hana-SS3_data_mnt00001
stlrx300s8-6:/ # mkfs.xfs /dev/mapper/hana-SS3_log_mnt00001
stlrx300s8-6:/ # mkfs.xfs /dev/mapper/hana-SS3_shared
```

- For a multiple-host system, create the XFS file system on all data and log LUNs.

```
stlrx300s8-6:~ # mkfs.xfs /dev/mapper/hana-SS3_log_mnt00001
stlrx300s8-6:~ # mkfs.xfs /dev/mapper/hana-SS3_log_mnt00002
stlrx300s8-6:~ # mkfs.xfs /dev/mapper/hana-SS3_data_mnt00001
stlrx300s8-6:~ # mkfs.xfs /dev/mapper/hana-SS3_data_mnt00002
```

- If LVM is used, create the XFS file system on all data and log logical volumes.

```
mkfs.xfs FC5_data_mnt00001-vol
mkfs.xfs FC5_data_mnt00002-vol
mkfs.xfs FC5_log_mnt00001-vol
mkfs.xfs FC5_log_mnt00002-vol
```

The multiple host example commands show a 2+1 multiple-host HANA system.

## Create mount points

To create the required mount point directories, take one of the following actions:

- For a single-host system, set permissions and create mount points on the database host.

```
stlrx300s8-6:/ # mkdir -p /hana/data/SS3/mnt00001
stlrx300s8-6:/ # mkdir -p /hana/log/SS3/mnt00001
stlrx300s8-6:/ # mkdir -p /hana/shared

stlrx300s8-6:/ # chmod -R 777 /hana/log/SS3
stlrx300s8-6:/ # chmod -R 777 /hana/data/SS3
stlrx300s8-6:/ # chmod 777 /hana/shared
```

- For a multiple-host system, set permissions and create mount points on all worker and standby hosts.

The example commands show a 2+1 multiple-host HANA system.

```
stlrx300s8-6:/ # mkdir -p /hana/data/SS3/mnt00001
stlrx300s8-6:/ # mkdir -p /hana/log/SS3/mnt00001
stlrx300s8-6:/ # mkdir -p /hana/data/SS3/mnt00002
stlrx300s8-6:/ # mkdir -p /hana/log/SS3/mnt00002
stlrx300s8-6:/ # mkdir -p /hana/shared

stlrx300s8-6:/ # chmod -R 777 /hana/log/SS3
stlrx300s8-6:/ # chmod -R 777 /hana/data/SS3
stlrx300s8-6:/ # chmod 777 /hana/shared
```

The same steps must be executed for a system configuration with Linux LVM.

## Mount file systems

To mount file systems during system boot using the `/etc/fstab` configuration file, complete the following steps:

1. Take one of the following actions:
   - For a single-host system, add the required file systems to the `/etc/fstab` configuration file.

     The XFS file systems for the data and log LUN must be mounted with the `relatime` and `inode64` mount options.

     ```
     stlrx300s8-6:/ # cat /etc/fstab
     /dev/mapper/hana-SS3_shared /hana/shared xfs defaults 0 0
     /dev/mapper/hana-SS3_log_mnt00001 /hana/log/SS3/mnt00001 xfs
     relatime,inode64 0 0
     /dev/mapper/hana-SS3_data_mnt00001 /hana/data/SS3/mnt00001 xfs
     relatime,inode64 0 0
     ```

If LVM is used, use the logical volume names for data and log.

```
# cat /etc/fstab
/dev/mapper/hana-FC5_shared /hana/shared xfs defaults 0 0
/dev/mapper/FC5_log_mnt00001-vol /hana/log/FC5/mnt00001 xfs
relatime,inode64 0 0
/dev/mapper/FC5_data_mnt00001-vol /hana/data/FC5/mnt00001 xfs
relatime,inode64 0 0
```

- For a multiple-host system, add the `/hana/shared` file system to the `/etc/fstab` configuration file of each host.

  All the data and log file systems are mounted through the SAP HANA storage connector.

```
stlrx300s8-6:/ # cat /etc/fstab
<storage-ip>:/hana_shared /hana/shared nfs
rw,vers=3,hard,timeo=600,intr,noatime,nolock 0 0
```

2. To mount the file systems, run the `mount -a` command at each host.

Next: I/O stack configuration for SAP HANA.

# I/O stack configuration for SAP HANA

Previous: Host setup.

Starting with SAP HANA 1.0 SPS10, SAP introduced parameters to adjust the I/O behavior and optimize the database for the file and storage system used.

NetApp conducted performance tests to define the ideal values. The following table lists the optimal values as inferred from the performance tests.

| Parameter | Value |
|---|---|
| max_parallel_io_requests | 128 |
| async_read_submit | on |
| async_write_submit_active | on |
| async_write_submit_blocks | all |

For SAP HANA 1.0 up to SPS12, these parameters can be set during the installation of the SAP HANA database as described in SAP Note 2267798 – Configuration of the SAP HANA Database during Installation Using hdbparam.

Alternatively, the parameters can be set after the SAP HANA database installation using the `hdbparam` framework.

```
SS3adm@stlrx300s8-6:/usr/sap/SS3/HDB00> hdbparam --paramset
fileio.max_parallel_io_requests=128
SS3adm@stlrx300s8-6:/usr/sap/SS3/HDB00> hdbparam --paramset
fileio.async_write_submit_active=on
SS3adm@stlrx300s8-6:/usr/sap/SS3/HDB00> hdbparam --paramset
fileio.async_read_submit=on
SS3adm@stlrx300s8-6:/usr/sap/SS3/HDB00> hdbparam --paramset
fileio.async_write_submit_blocks=all
```

Starting with SAP HANA 2.0, `hdbparam` is deprecated and the parameters have been moved to the `global.ini` file. The parameters can be set by using SQL commands or SAP HANA Studio. For more information, see SAP Note 2399079 - Elimination of hdbparam in HANA 2. The parameters can be also set within the `global.ini` file.

```
SS3adm@stlrx300s8-6:/usr/sap/SS3/SYS/global/hdb/custom/config> cat
global.ini
…
[fileio]
async_read_submit = on
async_write_submit_active = on
max_parallel_io_requests = 128
async_write_submit_blocks = all
…
```

With SAP HANA 2.0 SPS5 and later, you can use the `setParameter.py `script to set the parameters mentioned above.

```
fc5adm@sapcc-hana-tst-03:/usr/sap/FC5/HDB00/exe/python_support>
python setParameter.py
-set=SYSTEM/global.ini/fileio/max_parallel_io_requests=128
python setParameter.py -set=SYSTEM/global.ini/fileio/async_read_submit=on
python setParameter.py
-set=SYSTEM/global.ini/fileio/async_write_submit_active=on
python setParameter.py
-set=SYSTEM/global.ini/fileio/async_write_submit_blocks=all
```

Next: SAP HANA software installation.

# SAP HANA software installation

Previous: I/O stack configuration for SAP HANA.

## Install on single-host system

SAP HANA software installation does not require any additional preparation for a single-host system.

## Install on multiple-host system

> The following installation procedure is based on SAP HANA 1.0 SPS12 or later.

Before beginning the installation, create a `global.ini` file to enable use of the SAP storage connector during the installation process. The SAP storage connector mounts the required file systems at the worker hosts during the installation process. The `global.ini` file must be available in a file system that is accessible from all hosts, such as the `/hana/shared/SID` file system.

Before installing SAP HANA software on a multiple-host system, the following steps must be completed:

1. Add the following mount options for the data LUNs and the log LUNs to the `global.ini` file:
   - `relatime` and `inode64` for the data and log file system
2. Add the WWIDs of the data and log partitions. The WWIDs must match the alias names configured in the `/etc/multipath.conf` file.

   The following output shows an example of a 2+1 multiple-host setup in which the system identifier (SID) is SS3.

```
stlrx300s8-6:~ # cat /hana/shared/global.ini
[communication]
listeninterface = .global
[persistence]
basepath_datavolumes = /hana/data/SS3
basepath_logvolumes = /hana/log/SS3
[storage]
ha_provider = hdb_ha.fcClient
partition_*_*__prtype = 5
partition_*_data__mountoptions = -o relatime,inode64
partition_*_log__mountoptions = -o relatime,inode64,nobarrier
partition_1_data__wwid = hana-SS3_data_mnt00001
partition_1_log__wwid = hana-SS3_log_mnt00001
partition_2_data__wwid = hana-SS3_data_mnt00002
partition_2_log__wwid = hana-SS3_log_mnt00002
[system_information]
usage = custom
[trace]
ha_fcclient = info
stlrx300s8-6:~ #
```

   If LVM is used, the needed configuration is different. The example below shows a 2+1 multiple-host setup with SID=FC5.

```
sapcc-hana-tst-03:/hana/shared # cat global.ini
[communication]
listeninterface = .global
[persistence]
basepath_datavolumes = /hana/data/FC5
basepath_logvolumes = /hana/log/FC5
[storage]
ha_provider = hdb_ha.fcClientLVM
partition_*_*__prtype = 5
partition_*_data__mountOptions = -o relatime,inode64
partition_*_log__mountOptions = -o relatime,inode64
partition_1_data__lvmname = FC5_data_mnt00001-vol
partition_1_log__lvmname = FC5_log_mnt00001-vol
partition_2_data__lvmname = FC5_data_mnt00002-vol
partition_2_log__lvmname = FC5_log_mnt00002-vol
sapcc-hana-tst-03:/hana/shared #
```

Using the SAP `hdblcm` installation tool, start the installation by running the following command at one of the worker hosts. Use the `addhosts` option to add the second worker (sapcc-hana-tst-04) and the standby host (sapcc-hana-tst-05).
The directory where the prepared the `global.ini` file has been stored is included with the `storage_cfg` CLI option (`--storage_cfg=/hana/shared`).
Depending on the OS version being used, it might be necessary to install phyton 2.7 before installing the SAP HANA database.

```
sapcc-hana-tst-03:/mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/HDB_LCM_LINUX_X86_64 # ./hdblcm --action=install
--addhosts=sapcc-hana-tst-04:role=worker:storage_partion=2,sapcc-hana-tst
-05:role:=standby --storage_cfg=/hana(shared/shared


SAP HANA Lifecycle Management - SAP HANA Database 2.00.052.00.1599235305
************************************************************************



Scanning software locations...
Detected components:
    SAP HANA AFL (incl.PAL,BFL,OFL) (2.00.052.0000.1599259237) in
/mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/HDB_AFL_LINUX_X86_64/packages
    SAP HANA Database (2.00.052.00.1599235305) in /mnt/sapcc-
share/software/SAP/HANA2SP5-52/DATA_UNITS/HDB_SERVER_LINUX_X86_64/server
    SAP HANA Database Client (2.5.109.1598303414) in /mnt/sapcc-
share/software/SAP/HANA2SP5-52/DATA_UNITS/HDB_CLIENT_LINUX_X86_64/client
    SAP HANA Smart Data Access (2.00.5.000.0) in /mnt/sapcc-
```

```
share/software/SAP/HANA2SP5-
52/DATA_UNITS/SAP_HANA_SDA_20_LINUX_X86_64/packages
    SAP HANA Studio (2.3.54.000000) in /mnt/sapcc-
share/software/SAP/HANA2SP5-52/DATA_UNITS/HDB_STUDIO_LINUX_X86_64/studio
    SAP HANA Local Secure Store (2.4.24.0) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/HANA_LSS_24_LINUX_X86_64/packages
    SAP HANA XS Advanced Runtime (1.0.130.519) in /mnt/sapcc-
share/software/SAP/HANA2SP5-52/DATA_UNITS/XSA_RT_10_LINUX_X86_64/packages
    SAP HANA EML AFL (2.00.052.0000.1599259237) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/HDB_EML_AFL_10_LINUX_X86_64/packages
    SAP HANA EPM-MDS (2.00.052.0000.1599259237) in /mnt/sapcc-
share/software/SAP/HANA2SP5-52/DATA_UNITS/SAP_HANA_EPM-MDS_10/packages
    GUI for HALM for XSA (including product installer) Version 1 (1.014.1)
in /mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACALMPIUI14_1.zip
    XSAC FILEPROCESSOR 1.0 (1.000.85) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACFILEPROC00_85.zip
    SAP HANA tools for accessing catalog content, data preview, SQL
console, etc. (2.012.20341) in /mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSAC_HRTT_20/XSACHRTT12_20341.zip
    XS Messaging Service 1 (1.004.10) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACMESSSRV04_10.zip
    Develop and run portal services for customer apps on XSA (1.005.1) in
/mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACPORTALSERV05_1.zip
    SAP Web IDE Web Client (4.005.1) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSAC_SAP_WEB_IDE_20/XSACSAPWEBIDE05_1.zip
    XS JOB SCHEDULER 1.0 (1.007.12) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACSERVICES07_12.zip
    SAPUI5 FESV6 XSA 1 - SAPUI5 1.71 (1.071.25) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACUI5FESV671_25.zip
    SAPUI5 SERVICE BROKER XSA 1 - SAPUI5 Service Broker 1.0 (1.000.3) in
/mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACUI5SB00_3.zip
    XSA Cockpit 1 (1.001.17) in /mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACXSACOCKPIT01_17.zip


SAP HANA Database version '2.00.052.00.1599235305' will be installed.
```

```
Select additional components for installation:
  Index | Components | Description

-------------------------------------------------------------------------
-------------------
  1     | all        | All components
  2     | server     | No additional components
  3     | client     | Install SAP HANA Database Client version
2.5.109.1598303414
  4     | lss        | Install SAP HANA Local Secure Store version
2.4.24.0
  5     | studio     | Install SAP HANA Studio version 2.3.54.000000
  6     | smartda    | Install SAP HANA Smart Data Access version
2.00.5.000.0
  7     | xs         | Install SAP HANA XS Advanced Runtime version
1.0.130.519
  8     | afl        | Install SAP HANA AFL (incl.PAL,BFL,OFL) version
2.00.052.0000.1599259237
  9     | eml        | Install SAP HANA EML AFL version
2.00.052.0000.1599259237
  10    | epmmds     | Install SAP HANA EPM-MDS version
2.00.052.0000.1599259237

Enter comma-separated list of the selected indices [3]: 2,3
Enter Installation Path [/hana/shared]:
Enter Local Host Name [sapcc-hana-tst-03]:
```

Verify that the installation tool installed all selected components at all worker and standby hosts.

Next: Adding additional data volume partitions for SAP HANA single-host systems.

# Adding additional data volume partitions for SAP HANA single-host systems

Previous: SAP HANA software installation.

Starting with SAP HANA 2.0 SPS4, additional data volume partitions can be configured. This feature allows you to configure two or more LUNs for the data volume of an SAP HANA tenant database and to scale beyond the size and performance limits of a single LUN.

> It is not necessary to use multiple partitions to fulfil the SAP HANA KPIs. A single LUN with a single partition fulfils the required KPIs.

> Using two or more individual LUNs for the data volume is only available for SAP HANA single-host systems. The SAP storage connector required for SAP HANA multiple-host systems does only support one device for the data volume.

You can add more data volume partitions at any time but it might require a restart of the SAP HANA database.

## Enabling additional data volume partitions

To enable additional data volume partitions, complete the following steps:

1. Add the following entry within the `global.ini` file:

```
[customizable_functionalities]
persistence_datavolume_partition_multipath = true
```

2. Restart the database to enable the feature. Adding the parameter through the SAP HANA Studio to the `global.ini` file by using the Systemdb configuration prevents the restart of the database.

## Volume and LUN configuration

The layout of volumes and LUNs is similar to the layout of a single host with one data volume partition, but with an additional data volume and LUN stored on a different aggregate as log volume and the other data volume. The following table shows an example configuration of an SAP HANA single-host systems with two data volume partitions.

| Aggregate 1 at Controller A | Aggregate 2 at Controller A | Aggregate 1 at Controller B | Aggregate 2 at Controller B |
|---|---|---|---|
| Data volume: SID_data_mnt00001 | Shared volume: SID_shared | Data volume: SID_data2_mnt00001 | Log volume: SID_log_mnt00001 |

The next table shows an example of the mount point configuration for a single-host system with two data volume partitions.

| LUN | Mount point at HANA host | Note |
|---|---|---|
| SID_data_mnt00001 | /hana/data/SID/mnt00001 | Mounted using /etc/fstab entry |
| SID_data2_mnt00001 | /hana/data2/SID/mnt00001 | Mounted using /etc/fstab entry |
| SID_log_mnt00001 | /hana/log/SID/mnt00001 | Mounted using /etc/fstab entry |
| SID_shared | /hana/shared/SID | Mounted using /etc/fstab entry |

Create the new data LUNs by using either ONTAP System Manager or the ONTAP CLI.

## Host configuration

To configure a host, complete the following steps:

1. Configure multipathing for the additional LUNs, as described in section 0.
2. Create the XFS file system on each additional LUN belonging to the HANA system.

```
stlrx300s8-6:/ # mkfs.xfs /dev/mapper/hana-SS3_data2_mnt00001
```

3. Add the additional file system/s to the `/etc/fstab` configuration file.

> The XFS file systems for the data LUN must be mounted with the `relatime` and `inode64` mount options. The XFS file systems for the log LUN must be mounted with the `relatime`, `inode64`, and `nobarrier` mount options.

```
stlrx300s8-6:/ # cat /etc/fstab
/dev/mapper/hana-SS3_shared /hana/shared xfs defaults 0 0
/dev/mapper/hana-SS3_log_mnt00001 /hana/log/SS3/mnt00001 xfs
relatime,inode64 0 0
/dev/mapper/hana-SS3_data_mnt00001 /hana/data/SS3/mnt00001 xfs
relatime,inode64 0 0
/dev/mapper/hana-SS3_data2_mnt00001 /hana/data2/SS3/mnt00001 xfs
relatime,inode64 0 0
```

4. Create the mount points and set the permissions on the database host.

```
stlrx300s8-6:/ # mkdir -p /hana/data2/SS3/mnt00001
stlrx300s8-6:/ # chmod -R 777 /hana/data2/SS3
```

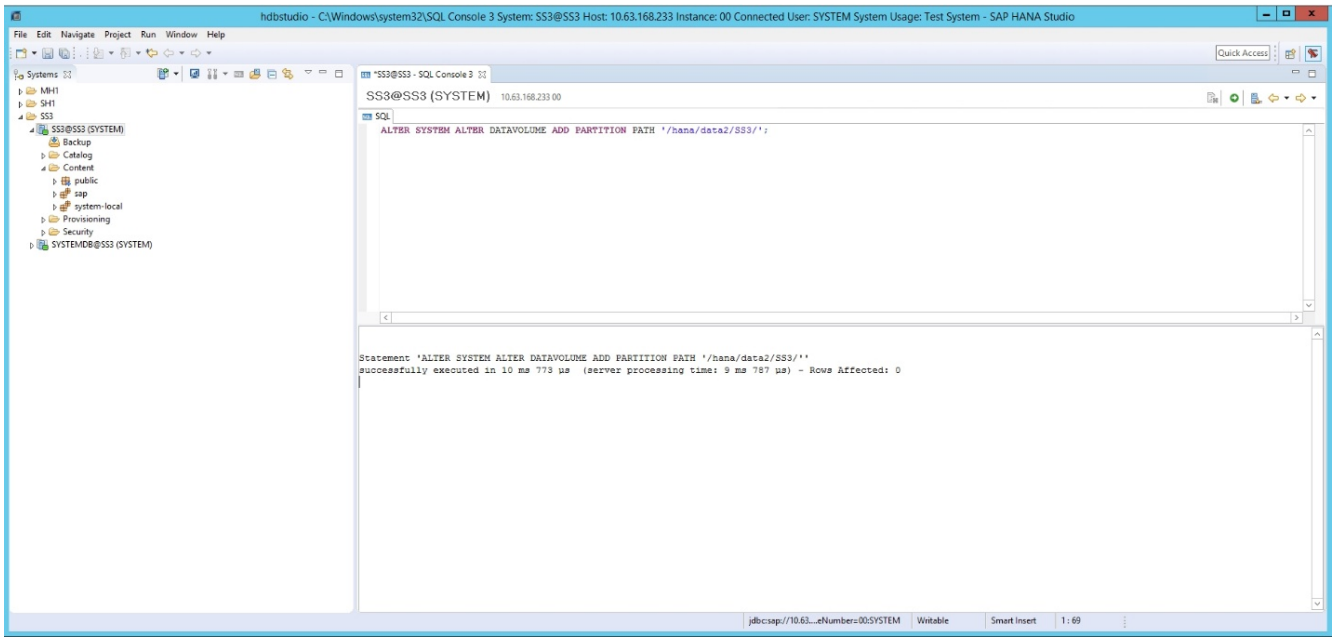
5. To mount the file systems, run the `mount -a` command.

## Adding an additional datavolume partition

To add an additional datavolume partition to your tenant database, complete the following step:

1. Execute the following SQL statement against the tenant database. Each additional LUN can have a different path.

```
ALTER SYSTEM ALTER DATAVOLUME ADD PARTITION PATH '/hana/data2/SID/';
```




Next: Where to find additional information.

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